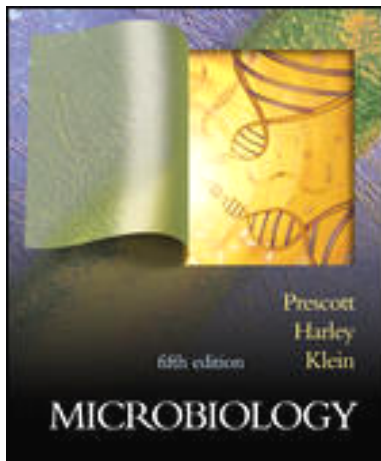




Microbiology

5th Edition

Lansing M. Prescott

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Overview

Prescott, Harley and Klein's 5th edition provides a balanced, comprehensive introduction to all major areas of microbiology. Because of this balance, *Microbiology, 5/e* is appropriate for students preparing for careers in medicine, dentistry, nursing, and allied health, as well as research, teaching, and industry. Biology and chemistry are prerequisites. The Fifth Edition has been updated extensively to reflect the latest discoveries in the field.

New to This Edition

- Every chapter in the book has been updated to reflect the latest discoveries in microbiology, including information on genomics, biofilms, mechanisms of toxins, classification, and emerging diseases. The most extensive revision has occurred in the areas of genetics, microbial ecology, and immunology where material has been updated and reorganized to allow for easier use.
- New Genomics chapter: Chapter 15. The genetics coverage has been reorganized for clarity and ease of teaching. The genetics section now ends with a completely new chapter on genomics. New Chapter 28 on microorganism interactions and microbial ecology!
- Newly developed art program--much of the art is new or revised! It incorporates color and style consistency throughout so students will easily identify certain topics.
- New critical thinking questions have been added to provide practice in analyzing data, predicting outcomes, and to teach students how to think logically.
- The general organization of the text has been modified to provide a more logical flow of topics and give greater emphasis to microbial ecology

Features

- Prescott's textbook contains briefer chapters than most books, but more of them (42). Students will find the concise chapters more palatable and less intimidating. Short chapters give the instructor the opportunity to fit the text more closely to the instructor's syllabus. Topic flexibility is allowed.
- There is an outstanding pedagogical system including outlines, concepts, key terms, cross-referencing, readings, new critical thinking questions, etc., to help students understand difficult material.

PREFACE

Books are the carriers of civilization. Without books, history is silent, literature dumb, science crippled, thought and speculation at a standstill. They are engines of change, windows on the world, lighthouses erected in a sea of time.

—Barbara Tuchman

Microbiology is an exceptionally broad discipline encompassing specialties as diverse as biochemistry, cell biology, genetics, taxonomy, pathogenic bacteriology, food and industrial microbiology, and ecology. A microbiologist must be acquainted with many biological disciplines and with all major groups of microorganisms: viruses, bacteria, fungi, algae, and protozoa. The key is balance. Students new to the subject need an introduction to the whole before concentrating on those parts of greatest interest to them. This text provides a balanced introduction to all major areas of microbiology for a variety of students. Because of this balance, the book is suitable for courses with orientations ranging from basic microbiology to medical and applied microbiology. Students preparing for careers in medicine, dentistry, nursing, and allied health professions will find the text just as useful as those aiming for careers in research, teaching, and industry. Two quarters/semesters each of biology and chemistry are assumed, and an overview of relevant chemistry is also provided in appendix I.

Organization and Approach

The book is organized flexibly so that chapters and topics may be arranged in almost any order. Each chapter has been made as self-contained as possible to promote this flexibility. Some topics are essential to microbiology and have been given more extensive treatment.

The book is divided into 11 parts. The first 6 parts introduce the foundations of microbiology: the development of microbiology, the structure of microorganisms, microbial growth and its control, metabolism, molecular biology and genetics, DNA technology and genomics, and the nature of viruses. Part Seven is a survey of the microbial world. In the fifth edition, the bacterial survey closely follows the general organization of the forthcoming second edition of *Bergey's Manual of Systematic Bacteriology*. Although principal attention is devoted to bacteria, eucaryotic microorganisms receive more than usual coverage. Fungi, algae, and protozoa are important in their own right. The introduction to their biology in chapters 25–27 is essential to understanding topics as diverse as clinical microbiology and microbial ecology. Part Eight focuses on the relationships of microorganisms to other organisms and the environment (microbial ecology). It also introduces aquatic and terrestrial microbiology. Chapter 28 presents the general principles underlying microbial ecology and environmental microbiology so that the subsequent chapters on aquatic and terrestrial habitats can be used without excessive redundancy. The chapter also describes various types

of microbial interactions such as mutualism, protocooperation, commensalism, and predation that occur in the environment. Parts Nine and Ten are concerned with pathogenicity, resistance, and disease. The three chapters in Part Nine describe normal microbiota, non-specific host resistance, the major aspects of the immune response, and medical immunology. Part Ten first covers such essential topics as microbial pathogenicity, antimicrobial chemotherapy, and epidemiology. Then chapters 38–40 survey the major human microbial diseases. The disease survey is primarily organized taxonomically on the chapter level; within each chapter diseases are covered according to their mode of transmission. This approach provides flexibility and allows the student easy access to information concerning any disease of interest. The survey is not a simple catalog of diseases; diseases are included because of their medical importance and their ability to illuminate the basic principles of disease and resistance. Part Eleven concludes the text with an introduction to food and industrial microbiology. Five appendices aid the student with a review of some basic chemical concepts and with extra information about important topics not completely covered in the text.

This text is designed to be an effective teaching tool. A text is only as easy for a student to use as it is easy to read. Readability has been enhanced by using a relatively simple, direct writing style, many section headings, and an organized outline format within each chapter. The level of difficulty has been carefully set with the target audience in mind. During preparation of the fifth edition, every sentence was carefully checked for clarity and revised when necessary. The American Society for Microbiology's *ASM Style Manual* conventions for nomenclature and abbreviations have been followed as consistently as possible.

The many new terms encountered in studying microbiology are a major stumbling block for students. This text lessens the problem by addressing and reinforcing a student's vocabulary development in three ways: (1) no new term is used without being clearly defined (often derivations also are given)—a student does not have to be familiar with the terminology of microbiology to use this text; (2) the most important terms are printed in boldface when first used; and (3) a very extensive, up-to-date, page-referenced glossary is included at the end of the text.

Because illustrations are critical to a student's learning and enjoyment of microbiology, all illustrations are full-color, and many excellent color photographs have been used. Color not only enhances the text's attractiveness but also increases each figure's teaching effectiveness. Considerable effort has gone into making the art as attractive and useful as possible. Much of the art in the

fourth edition has been revised and improved for use in the fifth edition. All new line art has been produced under the direct supervision of an art editor and the authors, and designed to illustrate and reinforce specific points in the text. Consequently every illustration is directly related to the narrative and specifically cited where appropriate. Great care has been taken to position illustrations as close as possible to the places where they are cited. Illustrations and captions have been reviewed for accuracy and clarity.

Themes in the Book

At least seven themes run through the text, though a particular one may be more obvious at some points than are others. These themes or emphases are the following:

1. The development of microbiology as a science
2. The nature and importance of the techniques used to isolate, culture, observe, and identify microorganisms
3. The control of microorganisms and reduction of their detrimental effects
4. The importance of molecular biology for microbiology
5. The medical significance of microbiology
6. The ways in which microorganisms interact with their environments and the practical consequences of these interactions
7. The influences that microorganisms and microbiological applications have on everyday life

These themes help unify the text and enhance continuity. The student should get a feeling for what microbiologists do and for how their activities affect society.

What's New in the Fifth Edition

Many substantial changes and improvements have been made in the fifth edition, including the following:

1. The general organization of the text has been modified to provide a more logical flow of topics and give greater emphasis to microbial ecology. Treatment of nucleic acid and protein synthesis has been moved to the genetics chapters to integrate the discussion of gene structure, replication, expression, and regulation. Recombinant DNA technology has been moved to a separate section, which also contains a new chapter on microbial genomics. The three-chapter introduction to microbial ecology now follows the survey of microbial diversity. This places it earlier in the text where basic principles of microbiology are introduced. Part Nine now contains a description of nonspecific host resistance as well as an introduction to the fundamentals of immunology. Symbiotic associations are discussed in the context of microbial ecology. The treatment of microbial pathogenesis has been expanded into a full chapter and placed with other medical topics in Part Ten.
2. Pedagogical aids have been expanded. A new Critical Thinking Questions section with two or more questions follows the Questions for Thought and Review. Section numbers have been given to all major chapter sections in

order to make cross references more precise. The summary now contains boldfaced references to tables and figures that will be useful in reviewing the chapter.

3. New illustrations have been added to almost every chapter. In addition, all figures have been carefully reviewed by our art editor, and many have been revised to improve their appearance and usefulness.
4. All reference sections have been revised and updated.

Besides these broader changes in the text, every chapter has been updated and often substantially revised. Some of the more important improvements are the following:

Chapter 1—A box on molecular Koch's postulates and a new section on the future of microbiology have been added.

Chapter 2—Differential interference contrast microscopy and confocal microscopy are described.

Chapter 3—More details on the mechanism of flagellar motion are provided.

Chapter 5—Phosphate uptake and ABC transporters are described.

Chapter 6—The chapter has new material on starvation proteins, growth limitation by environmental factors, viable but nonculturable procaryotes, and quorum sensing.

Chapter 8—The discussions of metabolic regulation and control of enzyme activity have been combined with the introduction to energy and enzymes.

Chapter 9—The metabolic overview has been rewritten to aid in understanding. The sections on electron transport, oxidative phosphorylation, and anaerobic respiration have been updated and expanded.

Chapter 11—The chapter now focuses on nucleic acid and gene structure, mutations, and DNA repair. New material on DNA methylation has been added.

Chapter 12—Material on gene expression (transcription and protein synthesis) has been moved here and combined with an extensive discussion of the regulation of gene expression. New sections on global regulatory systems and two-component phosphorelay systems have been added.

Chapter 15—This new chapter provides a brief introduction to microbial genomics, including genome sequencing, bioinformatics, general characteristics of microbial genomes, and functional genomics.

Chapter 18—Virus taxonomy has been updated and new life cycle diagrams added.

Chapter 19—Material on polyphasic taxonomy and the effects of horizontal gene transfer on phylogenetic trees has been added. The introduction to the second edition of *Bergey's Manual* has been revised and updated.

Chapters 20–24—The procaryotic survey chapters have been further revised to conform to the forthcoming second edition of *Bergey's Manual*.

Chapter 28—This chapter, formerly chapter 40, has been substantially rewritten and now includes a treatment of symbiosis and microbial interactions (e.g., mutualism, protozoosymbiosis, commensalism, predation, amensalism, competition, etc.). A discussion of microbial movement

between ecosystems has been added, and the treatment of biofilms and microbial mats has been expanded.

Chapter 29—The chapter on microorganisms in aquatic environments has new material on such topics as oxygen fluxes in water, the microbial loop, *Thiomargarita namibiensis*, microorganisms in freshwater ice, and current drinking water standards.

Chapter 30—Microorganisms in cold moist area soils, desert soils, and geologically heated hyperthermal soils are discussed. The effects of nitrogen, phosphorus, and atmospheric gases on plants and soils are described more extensively. There is a new section on the subsurface biosphere.

Chapter 31—This reorganized chapter discusses normal microbiota and nonspecific resistance. An overview of host resistance; a discussion of the cells, tissues, and organs of the immune system; an introduction to the alternative and lectin complement pathways; and a summary of cytokine properties and functions have been included.

Chapter 32—All aspects of specific immunity have been moved to this chapter in order to provide a clearer and more coherent discussion. The chapter contains an overview of specific immunity, a discussion of antigens and antibodies, T-cell and B-cell biology, a discussion of the action of antibodies, the classical complement pathway, and a section on acquired immune tolerance. It ends with a summary of the role of antibodies and lymphocytes in resistance.

Chapter 33—The new chapter on medical immunology contains topics more directly related to the practical aspects of health and clinical microbiology: vaccines and immunizations, immune disorders, and in vitro antigen-antibody interactions. Previously these were scattered over three chapters. The treatment of vaccines has been greatly expanded.

Chapter 34—The treatment of microbial pathogenicity has been greatly enlarged and made into a separate chapter. Several topics have been expanded or added: regulation of bacterial virulence factors and pathogenicity islands, the mechanisms of exotoxin action, and microbial mechanisms for escaping host defenses.

Chapter 37—In the epidemiology chapter, the treatment of emerging diseases has been expanded. New sections on bioterrorism and the effect of global travel on health have been added.

Chapters 38–40—The disease survey chapters have been brought up-to-date, and bacterial diseases are now covered in one chapter rather than two. New material has been added on genital herpes, listeriosis, the use of clostridial toxins in therapy, and other topics. A new table describing common sexually transmitted diseases and their treatment is provided.

Chapter 41—New aspects of food microbiology include discussions of modified atmosphere packaging, algal toxins, bacteriocins as preservatives, new variant Creutzfeldt-Jakob disease, food poisoning by uncooked foods, new techniques in tracing outbreaks of food-related diseases, and the use of probiotics in the diet.

Chapter 42—The chapter on industrial microbiology and biotechnology has been revised to include current advances

due to new molecular techniques. A section on developing and choosing microorganisms for use in industry has been added. Other topics that have been added or substantially revised include the synthesis of products for medical use, biodegradation of pesticides and other pollutants, the addition of microorganisms to the environment, and the use of microarray technology.

Aids to the Student

It is hard to overemphasize the importance of pedagogical aids for the student. Accuracy is most important, but if a text is not clear, readable, and attractive, up-to-dateness and accuracy are wasted because students will not read it. Students must be able to understand the material being presented, effectively use the text as a learning tool, and enjoy reading the book.

To be an effective teaching tool, a text must present the science of microbiology in a way that can be clearly taught and easily learned. Therefore many aids are included to make the task of learning more efficient and enjoyable. Following the preface a special section addressed to the student user reviews the principles of effective learning, including the SQ4R (survey, question, read, revise, record, and review) study technique. Specific chapter aids are described in the special Visual Preview section.

Besides the chapter aids the text also contains a glossary, an index, and five appendices. The extensive *glossary* defines the most important terms from each chapter and includes page references. Where desirable, phonetic pronunciations also are given. Most of the glossary definitions have not been taken directly from the text but have been rewritten to give the student further understanding of the item. To improve ease of use, the fifth edition has a large, detailed *index*. It has been carefully designed to make text material more accessible. The *appendices* aid the student with extra review of chemical principles and metabolic pathways and provide further details about the taxonomy of bacteria and viruses. To aid the student in following the rapidly changing field of procaryotic taxonomy, appendix III provides the classification of procaryotes according to the first edition of *Bergey's Manual of Systematic Bacteriology*, and appendix IV gives the classification used by the upcoming second edition of *Bergey's Manual*.

Supplementary Materials

Rich supplementary materials are available for students and instructors to assist learning and course management.

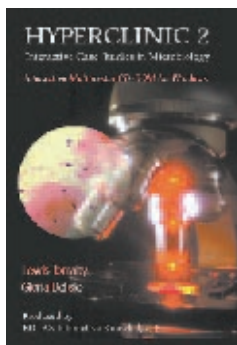
For the Student

1. A *Student Study Guide* by Linda Sherwood of Montana State University is a valuable resource that provides learning objectives, study outlines, learning activities, and self-testing material to help students master course content.
2. The *Interactive E-TEXT* available on CD-ROM in January 2002 includes all of *Microbiology*, Fifth Edition, as well as the



Student Study Guide in an interactive electronic format. The e-text includes animations and web links to enhance learning.

3. The third edition of *Microbes in Motion* by Gloria Delisle and Lewis Tomalty is an interactive CD-ROM that brings microbiology to life. A correlation guide on the CD links this exciting resource directly to your textbook. This easy to use tutorial can go from the classroom to the resource center to students' own personal computers. *Microbes in Motion* brings discovery back into the learning and education process through interactive screens, animations, video, audio, and hyperlinking questions. The applications of this CD-ROM are only as limited as your good ideas.
4. The second edition of *Hyperclinic* by Lewis Tomalty and Gloria Delisle is packed with over 100 case studies and over 200 pathogens supported with audio, video, and interactive screens. Students will have fun and gain confidence as they learn valuable concepts and gain practical experience in clinical microbiology.
5. The fifth edition of *Laboratory Exercises in Microbiology* by John P. Harley and Lansing M. Prescott has been prepared to accompany the text. Like the text, the laboratory manual provides a balanced introduction to laboratory techniques and principles that are important in each area of microbiology. The class-tested exercises are modular and short so that an instructor can easily choose only those exercises that fit his or her course. The fifth edition contains recipes for all reagents and media. New exercises in biotechnology have been added to this edition. A new appendix provides practice in solving dilution problems.
6. A set of 305 *Microbiology Study Cards* prepared by Kent M. Van De Graaff, F. Brent Johnson, Brigham Young University, and Christopher H. Creek features complete descriptions of terms, clearly labeled drawings, clinical information on diseases, and much more.



multimedia presentations or export images into other programs. Images may be sorted by a number of criteria. Features include an Interactive Slide Show and a Slide Editor.

4. A set of 50 *Projection Slides* provides clinical examples of diseases and pathogens to supplement the illustrations in the text.
5. Your McGraw-Hill representative may arrange a *Customized Laboratory Manual* combining your own material with exercises from *Laboratory Exercises in Microbiology*, Fifth Edition, by John P. Harley and Lansing M. Prescott. Contact your McGraw-Hill representative for details about this custom publishing service.
6. Designed specifically to help you with your individual course needs, *PageOut*, *PageOut Lite*, and *McGraw-Hill Course Solutions* will assist you in integrating your syllabus with the fifth edition's state-of-the-art media tools. Create your own course-specific web page supported by McGraw-Hill's extensive electronic resources, set up a class message board or chat room online, provide online testing opportunities for your students, and more!



Online Resources

Through the Prescott 2002 *Online Learning Center*, everything you need for effective, interactive teaching and learning is at your fingertips. Moreover, this vast McGraw-Hill resource is easily loaded into course management systems such as WebCT or Blackboard. Through the Online Learning Center, you will also link to McGraw-Hill's new *Biocourse.com* site with a huge dynamic array of resources to supplement your learning experience in microbiology.

Some of the online features you will find to support your use of *Microbiology* by Prescott, Harley, and Klein include the following.



For the Instructor

1. A *Testing CD* is offered free on request to adopters of the text. This cross-platform CD provides a database of over 2,500 objective questions for preparing exams and a grade-recording program.
2. A set of 250 full-color acetate *Transparencies* is available to supplement classroom lectures. These have been enhanced for projection and are available to adopters of the fifth edition.
3. The *Visual Resource Library* CD-ROM contains virtually all of the art and many of the photos from *Microbiology*, Fifth Edition, as well as the tables that appear in the text. This presentation software allows you to create your own

For the Student:

- Additional multiple-choice questions in a self-quizzing interactive format
- Electronic flashcards to review key vocabulary
- Study Outlines
- Web Links and Exercises
- Clinical Case Studies
- An Interactive Time Line detailing events and highlighting personalities critical to the development of microbiology
- Study Tips
- Student *Tutorial Service*

For the Instructor:

- A complete **Instructor's Manual and Test Item File** written by David Mullin of Tulane University. The Instructor's Manual contains chapter overviews and objectives, correlation guides, and more. The Test Item File containing over 2,500 questions, and password protected, provides a powerful instructional tool.
- The **Laboratory Resource Guide** provides answers to all exercises in *Laboratory Exercises in Microbiology*, Fifth Edition, by John P. Harley and Lansing M. Prescott.
- Images and tables from the text in a downloadable format for classroom presentation.
- **Correlation guides** for use of all resources available with the text and correlations of text material with the ASM Guidelines.
- **Answers to Critical Thinking Questions** in the text.
- **Web Links** to active microbiology sites and to other sites with teaching resources.
- A **Course Consultant** to answer your specific questions about using McGraw-Hill resources with your syllabus.

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classification would not have been possible without his assistance. We also much appreciate Amy Cheng Vollmer's contribution of critical thinking questions for each chapter. They will significantly enrich the student's learning experience. John Harley was greatly helped with the section on bioterrorism by James Snyder. Donald Klein wishes to acknowledge the aid of Jeffrey O. Dawson, Frank B. Dazzo, Arnold L. Demain, Frank G. Ethridge, Zoila R. Flores-Bustamente, Michael P. Shiaris, Donald B. Tait, and Jean K. Whelan.

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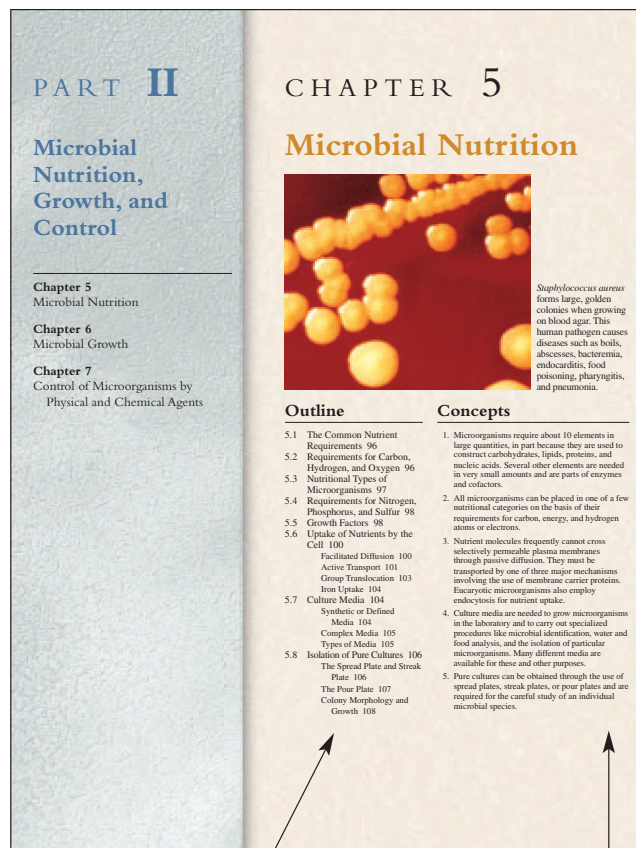
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John P. Harley
Donald A. Klein

VISUAL PREVIEW

The next few pages show you the tools found throughout the text to help you in your study of microbiology.

Opening Quotes are designed to perk student interest and provide perspective on chapter contents.

Chapter Preface is composed of one or two short paragraphs that preview the chapter contents and relate it to the rest of the text. The preface is not a summary, but allows the student to put the chapter into perspective at the start.



Chapter Outlines include all major headings in the chapter with section and page numbers. This helps the reader quickly locate topics of interest.

Chapter Concepts briefly summarize some of the most important concepts the student should master.

We all labour against our own cure, for death is the cure of all diseases.

—Sir Thomas Browne

The chapters in Part II are concerned with the nutrition, growth, and control of microorganisms. This chapter addresses the subject of the nonspecific control and destruction of microorganisms, a topic of immense practical importance. Although many microorganisms are beneficial and necessary for human well-being, microbial activities may have undesirable consequences, such as food spoilage and disease. Therefore it is essential to be able to kill a wide variety of microorganisms or inhibit their growth to minimize their destructive effects. The goal is twofold: (1) to destroy pathogens and prevent their transmission, and (2) to reduce or eliminate microorganisms responsible for the contamination of water, food, and other substances.

This chapter focuses on the control of microorganisms by non-specific physical and chemical agents. Chapter 35 introduces the use of antimicrobial chemotherapy to control microbial disease.

From the beginning of recorded history, people have practiced disinfection and sterilization, even though the existence of microorganisms was long unsuspected. The Egyptians used fire to sterilize infectious material and disinfectants to embalm bodies, and the Greeks burned sulfur to fumigate buildings. Mosaic law commanded the Hebrews to burn any clothing suspected of being contaminated with the leprosy bacterium. Today the ability to de-

7.1 Definition of Frequently Used Terms 137

stroy microorganisms is no less important; it makes possible the aseptic techniques used in microbiological research, the preservation of food, and the prevention of disease. The techniques described in this chapter are also essential to personal safety in both the laboratory and hospital (Box 7.1).

There are several ways to control microbial growth that have not been included in this chapter, but they should be considered for a more complete picture of how microorganisms are controlled. Chapter 6 describes the effects of osmotic activity, pH, temperature, O_2 , and radiation on microbial growth and survival (see pp. 121–31). Chapter 41 discusses the use of physical and chemical agents in food preservation (see pp. 600–60).

7.1 Definition of Frequently Used Terms

Terminology is especially important when the control of microorganisms is discussed because words like disinfectant and antiseptic often are used loosely. The situation is even more confusing because a particular treatment can either inhibit growth or kill depending on the conditions.

The ability to control microbial populations on inanimate objects, like eating utensils and surgical instruments, is of considerable practical importance. Sometimes it is necessary to eliminate all microorganisms from an object, whereas only partial destruction of the microbial population may be required in other situations. **Sterilization** [Latin *sterilis*, unable to produce offspring or barren] is the process by which all living cells, viable spores, viruses, and viroids (see chapter 18) are either destroyed or removed from an object or habitat. A sterile object is totally free of viable microorganisms, spores, and other infectious agents. When sterilization is achieved by a chemical agent, the chemical is called a sterilant. In

Box 7.1

Safety in the Microbiology Laboratory

Personal safety should be of major concern in all microbiology laboratories. It has been estimated that thousands of infections have been acquired in the laboratory, and many persons have died because of such infections. The two most common laboratory-acquired bacterial diseases are typhoid fever and brucellosis. Most deaths have come from typhoid fever (20 deaths) and Rocky Mountain spotted fever (13 deaths). Infections by fungi (histoplasmosis) and viruses (Venezuelan equine encephalitis and hepatitis B virus from monkeys) are also not uncommon. Hepatitis is the most frequently reported laboratory-acquired viral infection, especially in people working in clinical laboratories and with blood. In a survey of 426 U.S. hospital workers, 40% of those in clinical chemistry and 21% in microbiology had antibodies to hepatitis B virus, indicating their previous exposure (though only about 19% of these had disease symptoms).

Efforts have been made to determine the causes of these infections in order to enhance the development of better preventive measures. Although often it is not possible to determine the direct cause of infection,

some major potential hazards are clear. One of the most frequent causes of disease is the inhalation of an infectious aerosol. An aerosol is a gaseous suspension of liquid or solid particles that may be generated by accidents and laboratory operations such as spills, centrifuge accidents, removal of closures from shaken culture tubes, and plunging of contaminated loops into a flame. Accidents with hypodermic syringes and needles, such as self-inoculation and spraying solutions from the needle, also are common. Hypodermics should be employed only when necessary and then with care. Pipette accidents involving the mouth are another major source of infection; pipettes should be filled with the use of pipette aids and operated in such a way as to avoid creating aerosols.

People must exercise care and common sense when working with microorganisms. Operations that might generate infectious aerosols should be carried out in a biological safety cabinet. Bench tops and incubators should be disinfected regularly. Autoclaves must be maintained and operated properly to ensure adequate sterilization. Laboratory personnel should wash their hands thoroughly before and after finishing work.

Boxed Readings are found in most chapters and describe items of interest that are not essential to the primary thrust of the chapter. Topics include currently exciting research areas, the practical impact of microbial activities, items of medical significance, historical anecdotes, and descriptions of extraordinary organisms.

Additional Reading 151

Critical Thinking Questions

- Throughout history, spices have been used as preservatives and to cover up the taste of food that is slightly spoiled. The success of some spices led to a magical, ritualized use of many of them and possession of spices was often limited to priests or other powerful members of the community.
 - Choose a spice and trace its use geographically and historically. What is its common-day use today?
 - Spices grow and tend to be used predominantly in warmer climates. Explain.
- Design an experiment to determine whether an antimicrobial agent is acting as a cidal or static agent. How would you determine whether an agent is suitable for use as an antiseptic rather than as a disinfectant?
- Suppose that you are testing the effectiveness of disinfectants with the phenol coefficient test and obtained the following results:

Dilution	Bacterial Growth after Treatment		
	Disinfectant A	Disinfectant B	Disinfectant C
1/20	—	—	—
1/40	+	—	—
1/80	+	—	+
1/160	+	+	+
1/320	+	+	+

What disinfectant can you safely say is the most effective? Can you determine its phenol coefficient from these results?

General

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7.4 The Use of Physical Methods in Control

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7.5 The Use of Chemical Agents in Control

Belkin, S., Dahan, S., Levi, Y., and Tounti, D. 1999.

224 Chapter 10 Metabolism: The Use of Energy in Biosynthesis**Summary**

- In biosynthesis or anabolism, cells use energy to construct complex molecules from smaller, simpler precursors.
- Many important cell constituents are macromolecules, large polymers constructed of simple monomers.
- Although many catabolic and anabolic pathways share enzymes for the sake of efficiency, some of their enzymes are separate and independently regulated.
- Macromolecular components often undergo self-assembly to form the final molecule or complex.
- Photosynthetic CO₂ fixation is carried out by the Calvin cycle and may be divided into three phases: the carboxylation phase, the reduction phase, and the regeneration phase (figure 10.4). Three ATPs and two NADPHs are used during the incorporation of one CO₂.
- Glucogenesis is the synthesis of glucose and related sugars from nonglucose precursors.
- Glucose, fructose, and mannose are glucogenic intermediates or made directly from these: galactose is synthesized with nucleoside diphosphate derivatives. Bacteria and algae synthesize glucose and starch from adenosine diphosphate glucose.
- Phosphorus is obtained from inorganic or organic phosphate.

- Microorganisms can use cysteine, methionine, and inorganic sulfate as sulfur sources. Sulfate is reduced to sulfide during assimilatory sulfate reduction.
- Ammonia nitrogen can be directly assimilated by the activity of transaminases and either glutamate dehydrogenase or the glutamine synthetase-glutamate synthase system (figure 10.10-10.12).
- Nitrate is incorporated through assimilatory nitrate reduction catalyzed by the enzymes nitrate reductase and nitrite reductase.
- Nitrogen fixation is catalyzed by the nitrogenase complex. Atmospheric molecular nitrogen is reduced to ammonia, which is then incorporated into amino acids (figures 10.14 and 10.16).
- Amino acid biosynthetic pathways branch off from the central amphibolic pathways (figure 10.17).
- Anaplerotic reactions replace TCA cycle intermediates to keep the cycle in balance while it supplies biosynthetic precursors. Many anaplerotic enzymes catalyze CO₂ fixation reactions. The glyoxylate cycle is also anaplerotic.
- Purines and pyrimidines are nitrogenous bases found in DNA, RNA, and other molecules. The purine skeleton is synthesized beginning with ribose 5-phosphate and initially produces

inosinic acid. Pyrimidine biosynthesis starts with carbamoyl phosphate and aspartate, and ribose is added after the skeleton has been constructed.

Fatty acids are synthesized from acetyl-CoA, malonyl-CoA, and NADPH by the fatty acid synthetase system. During synthesis the intermediates are attached to the acyl carrier protein. Double bonds can be added in two different ways.

Triacylglycerols are made from fatty acids and glycerol phosphate. Phosphatidic acid is an important intermediate in this pathway.

Phospholipids like phosphatidylethanolamine can be synthesized from phosphatidic acid by forming CDP-diacylglycerol, then adding an amino acid.

Peptidoglycan synthesis is a complex process involving both UDP derivatives and the lipid carrier bacteraeolipid, which transports NAM-NAG-pentapeptide units across the cell membrane. Cross-links are formed by transpeptidation (figures 10.28 and 10.29).

Peptidoglycan synthesis occurs in discrete zones in the cell wall. Existing peptidoglycan is selectively degraded by autolysins so new material can be added.

Key Terms

acyl carrier protein (ACP) 220
adenine 217
anaplerotic reactions 216
assimilatory nitrate reduction 211
assimilatory sulfate reduction 210
autolysins 223
bacteraeolipid 221
Calvin cycle 207
carboxysomes 207
CO₂ fixation 216
cytosine 217
dissimilatory sulfate reduction 210
fatty acid 218
fatty acid synthetase 218
glucogenesis 209

glutamate dehydrogenase 211
glutamate synthase 211
glutamine synthetase 211
glyoxylate cycle 216
guanine 217
macromolecule 205
monomers 205
nitrate reductase 212
nitrite reductase 212
nitrogenase 213
nitrogen fixation 212
nucleoside 217
nucleotide 217
phosphate 210

phosphatidic acid 220
phosphoenolpyruvate 210
purine 216
pyrimidine 216
ribulose-1,5-bisphosphate carboxylase 208
self-assembly 207
transaminases 221
transpeptidation 223
triacylglycerol 220
tumor 205
uracil 217
uridine diphosphate glucose (UDPG) 209

Critical Thinking Questions contains questions designed to stimulate more analytical and synthetic reasoning.

Questions for Thought and Review at the end of the chapter contains factual questions and some thought-provoking questions to aid the student in reviewing, integrating, and applying the material in the chapter.

Additional Reading 225

Questions for Thought and Review

- Discuss the relationship between catabolism and anabolism. How does anabolism depend on catabolism?
- Suppose that a microorganism was growing on a medium that contained amino acids but no sugars. In general terms how would it synthesize the pentoses and hexoses it required?
- Activated carriers participate in carboxylation, lipid, and peptidoglycan synthesis. Briefly describe these carriers and their roles.
- Which two enzymes discussed in the chapter appear to be specific to the Calvin cycle?
- Why can phosphorus be directly incorporated into cell constituents whereas sulfur and nitrogen often cannot?
- What is unusual about the synthesis of peptides that takes place during peptidoglycan construction?

Critical Thinking Questions

- In metabolism important intermediates are covalently attached to carriers, as if to mark these as important so the cell does not lose track of them. Think about a hotel placing your room key on a very large ring. List a few examples of these carriers and indicate whether they are involved primarily in anabolism or catabolism.
- Intermediate carriers are in a limited supply—when they cannot be recycled because of a metabolic block, serious consequences ensue. Think of some examples of these consequences.

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10.4 **The Assimilation of Inorganic Phosphorus, Sulfur, and Nitrogen**

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Chapter Summaries are a series of brief numbered statements designed to serve more as a study guide than as a complete, detailed summary of the chapter. Useful tables and figures are cited in the summary.

Key Terms is a list of all boldfaced terms and is provided at the end of the chapter to emphasize the most significant facts and concepts. Each term is page-referenced to the page on which the term is first introduced in the chapter.

Additional Readings are provided for further study. Most are reviews, monographs, and *Scientific American* articles rather than original research papers. Publications cited in these reviews introduce sufficiently interested students to the research literature. References through early 2001 have been included. The reference sections are organized into topical groups that correspond to the major sections in each chapter. This arrangement provides ease of access for students interested in particular topics.

Review Questions appear in small boxes at the end of most major sections. These questions help the student master the section's factual material and major concepts before continuing with the chapter.

Numbered Headings identify each major topic and are used for easy reference throughout the text and the accompanying laboratory manual.

nucleosome. Thus DNA gently isolated from chromatin looks like a string of beads. The stretch of DNA between the beads or nucleosomes, the linker region, varies in length from 14 to over 100 base pairs. Histone H1 appears to associate with the linker regions to aid the folding of DNA into more complex chromatin structures (figure 11.9b). When folding reaches a maximum, the chromatin takes the shape of the visible chromosomes seen in eucaryotic cells during mitosis and meiosis (see figure 4.20).

1. What are nucleic acids? How do DNA and RNA differ in structure?
2. Describe in some detail the structure of the DNA double helix. What does it mean to say that the two strands are complementary and antiparallel?
3. What are histones and nucleosomes? Describe the way in which DNA is organized in the chromosomes of prokaryotes and eucaryotes.

11.3 DNA Replication

The replication of DNA is an extraordinarily important and complex process, one upon which all life depends. We shall first discuss the overall pattern of DNA synthesis and then examine the mechanism of DNA replication in greater depth.

Patterns of DNA Synthesis

Watson and Crick published their description of DNA structure in April 1953. Almost exactly one month later, a second paper appeared in which they suggested how DNA might be replicated. They hypothesized that the two strands of the double helix unwind from one another and separate (figure 11.10). Free nucleotides now line up along the two parental strands through complementary base pairing—A with T, G with C (figure 11.7). When these nucleotides are linked together by one or more enzymes, two replicates result, each containing a parental DNA strand and a newly formed strand. Research in subsequent years has proved Watson and Crick's hypothesis correct.

Replication patterns are somewhat different in prokaryotes and eucaryotes. For example, when the circular DNA chromosome of *E. coli* is copied, replication begins at a single point, the origin. Synthesis occurs at the **replication fork**, the place at which the DNA helix is unwound and individual strands are replicated. Two replication forks move outward from the origin until they have copied the whole **replicon**, that portion of the genome that contains an origin and is replicated as a unit. When the replication forks move around the circle, a structure shaped like the Greek letter theta (θ) is formed (figure 11.11). Finally, since the bacterial chromosome is a single replicon, the forks meet on the other side and two separate chromosomes are released.

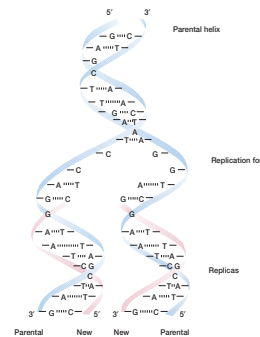


Figure 11.10 Semiconservative DNA Replication. The replication fork of DNA showing the synthesis of two progeny strands. Newly synthesized strands are in maroon. Each copy contains one new and one old strand. This process is called semiconservative replication.

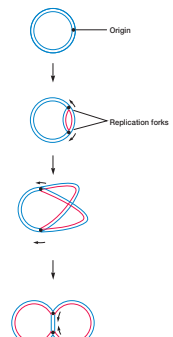


Figure 11.11 Bidirectional Replication. The replication of a circular bacterial genome. Two replication forks move around the DNA forming theta-shaped intermediates. Newly replicated DNA double helix is in red.

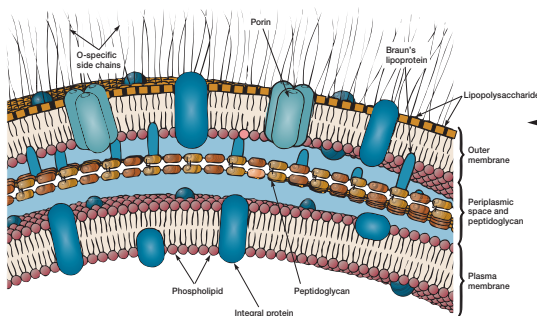


Figure 3.23 The Gram-Negative Envelope.

Multimedia-Supported Illustrations appear throughout the text. To facilitate finding corresponding full-color video, animations, or interactive screens from the third edition of *Microbes in Motion*, a correlation guide is provided on the CD-ROM, on the Student *Online Learning Center*, and in the *Student Study Guide*.

Microbes in Motion, Third Edition, CD-ROM is organized into 18 topical "books," the books are divided into "chapters," and the chapters have numbered "pages." For each multimedia-supported illustration, the correlation guide directs the reader to the book, chapter, and page on the CD-ROM where corresponding material can be found.

Figure 3.23 Bacterial Structure and Function Book Cell Wall Chapter Peptidoglycan Topic pp. 2–3

Cross-Reference Notes refer the student to major topics that are difficult and may need review in order to understand the current material. They also point the student either forward or backward to a related item of unusual interest or importance. Normally a reference is to either a specific section number or a page so that students can easily locate the item.

Boldfaced Terms are the important terms and are emphasized and clearly defined when they are first used. Bold terms are listed at the end of the chapter and most appear in the glossary.

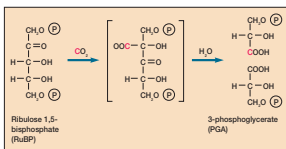


Figure 10.3 The Ribulose-1,5-bisphosphate Carboxylase Reaction. This enzyme catalyzes the addition of carbon dioxide to ribulose 1,5-bisphosphate, forming an unstable intermediate, which then breaks down to two molecules of 3-phosphoglycerate.

reduction, and regeneration. An overview of the cycle is given in figure 10.4 and the details are presented in appendix II.

The Carboxylation Phase

Carbon dioxide fixation is accomplished by the enzyme **ribulose 1,5-bisphosphate carboxylase** or **ribulosebiphosphate carboxylase/oxygenase** (rubisco) (figure 10.3), which catalyzes the addition of CO₂ to ribulose 1,5-bisphosphate (RuBP), forming two molecules of 3-phosphoglycerate (PGA).

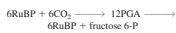
The Reduction Phase

After PGA is formed by carboxylation, it is reduced to glyceraldehyde 3-phosphate. The reduction, carried out by two enzymes, is essentially a reversal of a portion of the glycolytic pathway, although the glyceraldehyde 3-phosphate dehydrogenase differs from the glycolytic enzyme in using NADP⁺ rather than NAD⁺ (figure 10.4).

The Regeneration Phase

The third phase of the Calvin cycle regenerates RuBP and produces carbohydrates such as glyceraldehyde 3-phosphate, fructose, and glucose (figure 10.4). This portion of the cycle is similar to the pentose phosphate pathway and involves the transketolase and transaldolase reactions. The cycle is completed when phosphoribulokinase reforms RuBP.

To synthesize fructose 6-phosphate or glucose 6-phosphate from CO₂, the cycle must operate six times to yield the desired hexose and reform the six RuBP molecules.



The incorporation of one CO₂ into organic material requires three ATPs and two NADPHs. The formation of glucose from CO₂ may be summarized by the following equation.

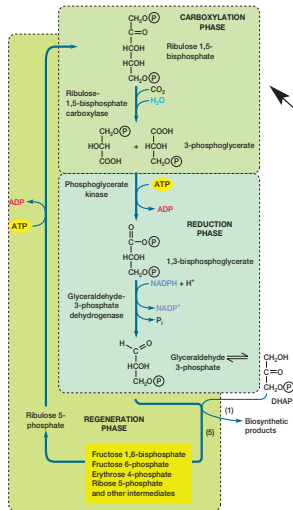
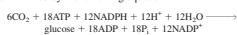


Figure 10.4 The Calvin Cycle. This is an overview of the cycle with only the carboxylation and reduction phases in detail. Three ribulose 1,5-bisphosphates are carboxylated to give six 3-phosphoglycerates in the carboxylation phase. These are converted to six glyceraldehyde 3-phosphates, which can be converted to dihydroxyacetone phosphate (DHAP). Five of the six trioses (glyceraldehyde phosphate and dihydroxyacetone phosphate) are used to reform three ribulose 1,5-bisphosphates in the regeneration phase. The remaining triose is used in biosynthesis.

ATP and NADPH are provided by photosynthetic light reactions or by oxidation of inorganic molecules in chemolithotrophs. Sugars formed in the Calvin cycle can then be used to synthesize other essential molecules.

rally occurring organic molecule that cannot be used by some microorganism. Actinomycetes will degrade any alcohol, paraffin, and even rubber. Some bacteria seem able to employ almost anything as a carbon source; for example, *Burkholderia cepacia* can use over 100 different carbon compounds. In contrast to these bacterial omnivores, some bacteria are exceedingly fastidious and catabolize only a few carbon compounds. Cultures of methylophilic bacteria metabolize methane, methanol, carbon monoxide, formic acid, and related one-carbon molecules. Parasitic members of the genus *Leptospira* use only long-chain fatty acids as their major source of carbon and energy.

It appears that in natural environments complex populations of microorganisms often will metabolize even relatively indigestible human-made substances such as pesticides. Indigestible molecules sometimes are oxidized and degraded in the presence of a growth-promoting nutrient that is metabolized at the same time, a process called cometabolism. The products of this breakdown process can then be used as nutrients by other microorganisms. (Degradation and microorganisms (pp. 000-000))

5.3 Nutritional Types of Microorganisms

In addition to the need for carbon, hydrogen, and oxygen, all organisms require sources of energy and electrons for growth to take place. Microorganisms can be grouped into nutritional classes based on how they satisfy all these requirements (table 5.1). We have already seen that microorganisms can be classified as either heterotrophs or autotrophs with respect to their preferred source of carbon. There are only two sources of energy available to organisms: (1) light energy, and (2) the energy derived from oxidizing organic or inorganic molecules. **Phototrophs** use light as their energy source; **chemotrophs** obtain energy from the oxidation of chemical compounds (either organic or inorganic). Microorganisms also have only two sources for

electrons. **Lithotrophs** (i.e., "rock-eaters") use reduced inorganic substances as their electron source, whereas **organotrophs** extract electrons from organic compounds. (Photosynthesis light reaction (pp. 195-201); Oxidation of organic and inorganic molecules (pp. 176-195))

Despite the great metabolic diversity seen in microorganisms, most may be placed in one of four nutritional classes based on their primary sources of carbon, energy, and electrons (table 5.2). The large majority of microorganisms thus far studied are either photolithotrophic autotrophs or chemolithotrophic heterotrophs. **Photolithotrophic autotrophs** (often called **photoautotrophs** or **photolithotrophs**) use light energy and have CO₂ as their carbon source. Eucaryotic algae and cyanobacteria employ water as the electron donor and release oxygen. Purple and green sulfur

Table 5.1 Sources of Carbon, Energy, and Electrons

Carbon Sources	Energy Sources	Electron Sources
Autotrophs	CO ₂ sole or principal biosynthetic carbon source (pp. 205-40)	Reduced inorganic molecules (pp. 193-94)
Heterotrophs	Reduced, preformed, organic molecules from other organisms (chapters 9 and 10)	Organic molecules (chapter 9)
Phototrophs	Light (pp. 195-201)	
Chemotrophs	Oxidation of organic or inorganic compounds (chapter 9)	

*For each category, the location of material describing the participating metabolic pathways is given within the parentheses.

Table 5.2 Major Nutritional Types of Microorganisms

Major Nutritional Type*	Sources of Energy, Hydrogen/Electrons, and Carbon	Representative Microorganisms
Photolithotrophic autotrophy (Photolithotrophy)	Light energy Inorganic hydrogen/electron (H ₂) donor CO ₂ carbon source	Algae Purple and green sulfur bacteria Cyanobacteria
Photoorganotrophic heterotrophy (Photoorganotrophy)	Light energy Organic H ₂ /e ⁻ donor Organic carbon source (CO ₂ may also be used)	Purple nonsulfur bacteria Green nonsulfur bacteria
Chemolithotrophic autotrophy (Chemolithotrophy)	Chemical energy source (inorganic) Inorganic H ₂ /e ⁻ donor CO ₂ carbon source	Sulfur-oxidizing bacteria Hydrogen bacteria Nitrifying bacteria Iron-oxidizing bacteria Protonema Fungi
Chemorganotrophic heterotrophy (Chemorganotrophy)	Chemical energy source (organic) Organic H ₂ /e ⁻ donor Organic carbon source	Most nonphotosynthetic bacteria (including most pathogens)

*Bacteria in other nutritional categories have been found. The categories are defined in terms of energy, electron, and carbon sources. Condensed versions of these names are given in parentheses.

New Figures and Tables have been added to this edition that summarize complex information in a concise presentation.

All Figure and Table References appear in bold type within the text for easy correlation between text and visual support elements.

TO THE STUDENT

One of the most important factors contributing to success in college, and in microbiology courses, is the use of good study techniques. This textbook is organized to help you to study more efficiently. But even a text with many learning aids is not effective unless used properly. Thus this section briefly outlines some practical study skills that will help ensure success in microbiology and make your use of this textbook more productive. Many of you already have the study skills mentioned here and will not need to spend time reviewing familiar material. These suggestions are made in the hope that they may be useful to those who are unaware of approaches like the SQ4R technique for studying textbooks.

Time Management and Study Environment

Many students find it difficult to study effectively because of a lack of time management and a proper place to study. Often a student will do poorly in courses because not enough time has been spent studying outside class. For best results you should plan to spend at least an average of four to eight hours a week outside class working on each course. There is sufficient time in the week for this, but it does require time management. If you spend a few minutes early in the morning planning how the day is to be used and allow adequate time for studying, much more will be accomplished. Students who make efficient use of every moment find that they have plenty of time for recreation.

A second important factor is a proper place to study so that you can concentrate and efficiently use your study time. Try to find a quiet location with a desk and adequate lighting. If possible, always study in the same place and use it only for studying. In this way you will be mentally prepared to study when you are at your desk. This location may be in the dorm, the library, a special study room, or somewhere else. Wherever it is, your study area should be free from distractions—including friends who drop by to socialize. Much more will be accomplished if you really study during your designated study times.

Making the Most of Lectures

Attendance at lectures is essential for success. Students who chronically miss classes usually do not do well. To gain the most from lectures, it is best to read any relevant text material beforehand. Be prepared to concentrate during lectures; do not simply sit back passively and listen to the instructor. During the lecture record your notes in a legible way so that you can understand them later. It is most efficient to employ an outline or simple paragraph format. The use of abbreviations or some type of shorthand notation often is effective. During lecture concentrate on what is being said and be sure to capture all of the main ideas, concepts, and definitions of important terms. Do not take sketchy notes assuming that you will remember things because

they are easy or obvious; you won't. Diagrams, lists, and terms written on the board are almost always important, as is anything the instructor clearly emphasizes by tone of voice. Feel free to ask questions during class when you don't understand something or wish the instructor to pursue a point further. Remember that if you don't understand, it is very likely that others in the class don't either but simply aren't willing to show their confusion. As soon as possible after a lecture, carefully review your notes to be certain that they are complete and understandable. Refer to the textbook when uncertain about something in your notes; it will be invaluable in clearing up questions and amplifying major points. When studying your notes for tests, it is a good idea to emphasize the most important points with a highlighter just as you would when reading the textbook.

Studying the Textbook

Your textbook is one of the most important learning tools in any course and should be very carefully and conscientiously used. Many years ago Francis P. Robinson developed a very effective study technique called SQ3R (survey, question, read, recite, and review). More recently L. L. Thistlethwaite and N. K. Snouffer have slightly modified it to yield the SQ4R approach (survey, question, read, revise, record, and review). This latter approach is summarized here:

1. *Survey.* Briefly scan the chapter to become familiar with its general content. Quickly read the title, introduction, summary, and main headings. Record the major ideas and points that you think the chapter will make. If there are a list of chapter concepts and a chapter outline, pay close attention to these. This survey should give you a feel for the topic and how the chapter is approaching it.
2. *Question.* As you reach each main heading or subheading, try to compose an important question or two that you believe the section will answer. This preview question will help focus your reading of the section. It is also a good idea to keep asking yourself questions as you read. This habit facilitates active reading and learning.
3. *Read.* Carefully read the section. Read to understand concepts and major points, and try to find the answer to your preview question(s). You may want to highlight very important terms or explanations of concepts, but do not indiscriminantly highlight everything. Be sure to pay close attention to any terms printed in color or boldface since the author(s) considered these to be important.
4. *Revise.* After reading the section, revise your question(s) to more accurately reflect the section's contents. These questions should be concept type questions that force you to bring together a number of details. They can be written in the margins of your text.

5. *Record.* Underline the information in the text that answers your questions, if you have not already done so. You may wish to write down the answers in note form as well. This process will give you good material to use in preparing for exams.
6. *Review.* Review the information by trying to answer your questions without looking at the text. If the text has a list of key words and a set of study questions, be sure to use these in your review. You will retain much more if you review the material several times.

Preparing for Examinations

It is extremely important to prepare for examinations properly so that you will not be rushed and tired on examination day. All textbook reading and lecture note revision should be completed

well ahead of time so that the last few days can be spent in mastering the material, not in trying to understand the basic concepts. Cramming at the last moment for an exam is no substitute for daily preparation and review. By managing time carefully and keeping up with your studies, you will have plenty of time to review thoroughly and clear up any questions. This will allow you to get sufficient rest before the test and to feel confident in your preparation. Because both physical condition and general attitude are important factors in test performance, you will automatically do better. Proper reviewing techniques also aid retention of the material.

Our website (www.mhhe.com/prescott5) contains many useful study aids. For example, the Student Center has more study tips, chapter overviews and outlines with links, flash cards, quizzes, a tutorial service, microbiology web links, clinical case studies, a Microbiology in the News page, and a correlation guide to the Microbes in Motion program.

For more useful study aids visit www.mhhe.com/prescott5.

PART I

Introduction to Microbiology

Chapter 1

The History and Scope
of Microbiology

Chapter 2

The Study of Microbial Structure:
Microscopy and Specimen
Preparation

Chapter 3

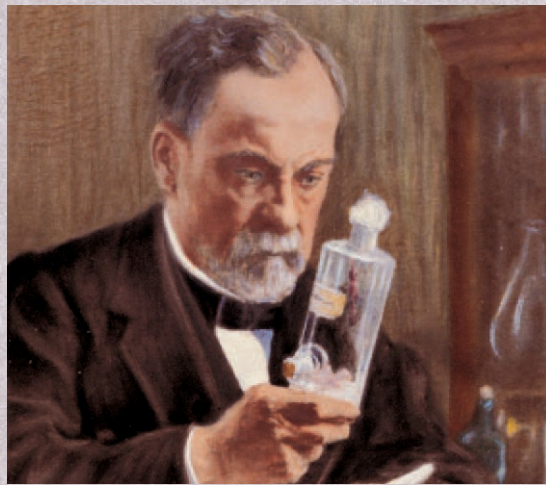
Prokaryotic Cell Structure
and Function

Chapter 4

Eukaryotic Cell Structure
and Function

CHAPTER 1

The History and Scope of Microbiology



Louis Pasteur, one of the greatest scientists of the nineteenth century, maintained that "Science knows no country, because knowledge belongs to humanity, and is a torch which illuminates the world."

Outline

- 1.1 The Discovery of Microorganisms 2
- 1.2 The Conflict over Spontaneous Generation 2
- 1.3 The Role of Microorganisms in Disease 7
 - Recognition of the Relationship between Microorganisms and Disease 7
 - The Development of Techniques for Studying Microbial Pathogens 8
 - Immunological Studies 9
- 1.4 Industrial Microbiology and Microbial Ecology 10
- 1.5 Members of the Microbial World 11
- 1.6 The Scope and Relevance of Microbiology 11
- 1.7 The Future of Microbiology 13

Concepts

1. Microbiology is the study of organisms that are usually too small to be seen by the unaided eye; it employs techniques—such as sterilization and the use of culture media—that are required to isolate and grow these microorganisms.
2. Microorganisms are not spontaneously generated from inanimate matter but arise from other microorganisms.
3. Many diseases result from viral, bacterial, fungal, or protozoan infections. Koch's postulates may be used to establish a causal link between the suspected microorganism and a disease.
4. The development of microbiology as a scientific discipline has depended on the availability of the microscope and the ability to isolate and grow pure cultures of microorganisms.
5. Microorganisms are responsible for many of the changes observed in organic and inorganic matter (e.g., fermentation and the carbon, nitrogen, and sulfur cycles that occur in nature).
6. Microorganisms have two fundamentally different types of cells—prokaryotic and eukaryotic—and are distributed among several kingdoms or domains.
7. Microbiology is a large discipline, which has a great impact on other areas of biology and general human welfare.

Dans les champs de l'observation, le hasard ne favorise que les esprits préparés.

(In the field of observation, chance favors only prepared minds.)

—Louis Pasteur

One can't overemphasize the importance of microbiology. Society benefits from microorganisms in many ways. They are necessary for the production of bread, cheese, beer, antibiotics, vaccines, vitamins, enzymes, and many other important products. Indeed, modern biotechnology rests upon a microbiological foundation. Microorganisms are indispensable components of our ecosystem. They make possible the cycles of carbon, oxygen, nitrogen, and sulfur that take place in terrestrial and aquatic systems. They also are a source of nutrients at the base of all ecological food chains and webs.

Of course microorganisms also have harmed humans and disrupted society over the millennia. Microbial diseases undoubtedly played a major role in historical events such as the decline of the Roman Empire and the conquest of the New World. In 1347 plague or black death (*see chapter 39*) struck Europe with brutal force. By 1351, only four years later, the plague had killed 1/3 of the population (about 25 million people). Over the next 80 years, the disease struck again and again, eventually wiping out 75% of the European population. Some historians believe that this disaster changed European culture and prepared the way for the Renaissance. Today the struggle by microbiologists and others against killers like AIDS and malaria continues. [The biology of AIDS and its impact \(pp. 878–84\)](#)

In this introductory chapter the historical development of the science of microbiology is described, and its relationship to medicine and other areas of biology is considered. The nature of the microbial world is then surveyed to provide a general idea of the organisms and agents that microbiologists study. Finally, the scope, relevance, and future of modern microbiology are discussed.

Microbiology often has been defined as the study of organisms and agents too small to be seen clearly by the unaided eye—that is, the study of **microorganisms**. Because objects less than about one millimeter in diameter cannot be seen clearly and must be examined with a microscope, microbiology is concerned primarily with organisms and agents this small and smaller. Its subjects are viruses, bacteria, many algae and fungi, and protozoa (*see table 34.1*). Yet other members of these groups, particularly some of the algae and fungi, are larger and quite visible. For example, bread molds and filamentous algae are studied by microbiologists, yet are visible to the naked eye. Two bacteria that are visible without a microscope, *Thiomargarita* and *Epulopiscium*, also have been discovered (*see p. 45*). The difficulty in setting the boundaries of microbiology led Roger Stanier to suggest that the field be defined not only in terms of the size of its subjects but also in terms

of its techniques. A microbiologist usually first isolates a specific microorganism from a population and then cultures it. Thus microbiology employs techniques—such as sterilization and the use of culture media—that are necessary for successful isolation and growth of microorganisms.

The development of microbiology as a science is described in the following sections. **Table 1.1** presents a summary of some of the major events in this process and their relationship to other historical landmarks.

1.1 The Discovery of Microorganisms

Even before microorganisms were seen, some investigators suspected their existence and responsibility for disease. Among others, the Roman philosopher Lucretius (about 98–55 B.C.) and the physician Girolamo Fracastoro (1478–1553) suggested that disease was caused by invisible living creatures. The earliest microscopic observations appear to have been made between 1625 and 1630 on bees and weevils by the Italian Francesco Stelluti, using a microscope probably supplied by Galileo. However, the first person to observe and describe microorganisms accurately was the amateur microscopist Antony van Leeuwenhoek (1632–1723) of Delft, Holland (**figure 1.1a**). Leeuwenhoek earned his living as a draper and haberdasher (a dealer in men's clothing and accessories), but spent much of his spare time constructing simple microscopes composed of double convex glass lenses held between two silver plates (*figure 1.1b*). His microscopes could magnify around 50 to 300 times, and he may have illuminated his liquid specimens by placing them between two pieces of glass and shining light on them at a 45° angle to the specimen plane. This would have provided a form of dark-field illumination (*see chapter 2*) and made bacteria clearly visible (*figure 1.1c*). Beginning in 1673 Leeuwenhoek sent detailed letters describing his discoveries to the Royal Society of London. It is clear from his descriptions that he saw both bacteria and protozoa.

1.2 The Conflict over Spontaneous Generation

From earliest times, people had believed in **spontaneous generation**—that living organisms could develop from nonliving matter. Even the great Aristotle (384–322 B.C.) thought some of the simpler invertebrates could arise by spontaneous generation. This view finally was challenged by the Italian physician Francesco Redi (1626–1697), who carried out a series of experiments on decaying meat and its ability to produce maggots spontaneously. Redi placed meat in three containers. One was uncovered, a second was covered with paper, and the third was covered with a fine gauze that would exclude flies. Flies laid their eggs on the uncovered meat and maggots developed. The other two pieces of meat did not produce maggots spontaneously. However, flies were attracted to the gauze-covered container and laid their eggs on the gauze; these eggs produced maggots.

Table 1.1 Some Important Events in the Development of Microbiology

Date	Microbiological History	Other Historical Events
1546	Fracastoro suggests that invisible organisms cause disease	Publication of Copernicus's work on the heliocentric solar system (1543)
1590–1608	Jansen develops first useful compound microscope	Shakespeare's <i>Hamlet</i> (1600–1601)
1676	Leeuwenhoek discovers "animalcules"	J. S. Bach and Handel born (1685)
1688	Redi publishes work on spontaneous generation of maggots	Isaac Newton publishes the <i>Principia</i> (1687)
		Linnaeus's <i>Systema Naturae</i> (1735)
		Mozart born (1756)
1765–1776	Spallanzani attacks spontaneous generation	
1786	Müller produces first classification of bacteria	French Revolution (1789)
1798	Jenner introduces cowpox vaccination for smallpox	Beethoven's first symphony (1800)
		The battle of Waterloo and the defeat of Napoleon (1815)
		Faraday demonstrates the principle of an electric motor (1821)
		England issues first postage stamp (1840)
1838–1839	Schwann and Schleiden, the Cell Theory	
1835–1844	Bassi discovers that silkworm disease is caused by a fungus and proposes that many diseases are microbial in origin	Marx's <i>Communist Manifesto</i> (1848)
1847–1850	Semmelweis shows that childbed fever is transmitted by physicians and introduces the use of antiseptics to prevent the disease	Velocity of light first measured by Fizeau (1849)
1849	Snow studies the epidemiology of a cholera epidemic in London	Clausius states the first and second laws of thermodynamics (1850)
		Graham distinguishes between colloids and crystalloids
		Melville's <i>Moby Dick</i> (1851)
		Otis installs first safe elevator (1854)
		Bunsen introduces the use of the gas burner (1855)
1857	Pasteur shows that lactic acid fermentation is due to a microorganism	
1858	Virchow states that all cells come from cells	Darwin's <i>On the Origin of Species</i> (1859)
1861	Pasteur shows that microorganisms do not arise by spontaneous generation	American Civil War (1861–1865)
		Mendel publishes his genetics experiments (1865)
		Cross-Atlantic cable laid (1865)
1867	Lister publishes his work on antiseptic surgery	Dostoevski's <i>Crime and Punishment</i> (1866)
1869	Miescher discovers nucleic acids	Franco-German War (1870–1871)
1876–1877	Koch demonstrates that anthrax is caused by <i>Bacillus anthracis</i>	Bell invents telephone (1876)
1880	Laveran discovers <i>Plasmodium</i> , the cause of malaria	Edison's first light bulb (1879)
1881	Koch cultures bacteria on gelatin	
	Pasteur develops anthrax vaccine	Ives produces first color photograph (1881)
1882	Koch discovers tubercle bacillus, <i>Mycobacterium tuberculosis</i>	
1884	Koch's postulates first published	First central electric power station constructed by Edison (1882)
	Metchnikoff describes phagocytosis	Mark Twain's <i>The Adventures of Huckleberry Finn</i> (1884)
	Autoclave developed	
	Gram stain developed	
1885	Pasteur develops rabies vaccine	
	Escherich discovers <i>Escherichia coli</i> , a cause of diarrhea	First motor vehicles developed by Daimler (1885–1886)
1886	Fraenkel discovers <i>Streptococcus pneumoniae</i> , a cause of pneumonia	
1887	Petri dish (plate) developed by Richard Petri	
1887–1890	Winogradsky studies sulfur and nitrifying bacteria	
1889	Beijerinck isolates root nodule bacteria	Hertz discovers radio waves (1888)
1890	Von Behring prepares antitoxins for diphtheria and tetanus	Eastman makes box camera (1888)
1892	Ivanowsky provides evidence for virus causation of tobacco mosaic disease	
		First zipper patented (1895)
1894	Kitasato and Yersin discover <i>Yersinia pestis</i> , the cause of plague	
1895	Bordet discovers complement	Röntgen discovers X rays (1895)
1896	Van Ermengem discovers <i>Clostridium botulinum</i> , the cause of botulism	
1897	Buchner prepares extract of yeast that ferments	Thomson discovers the electron (1897)
	Ross shows that malaria parasite is carried by the mosquito	Spanish-American War (1898)
1899	Beijerinck proves that a virus particle causes the tobacco mosaic disease	
1900	Reed proves that yellow fever is transmitted by the mosquito	Planck develops the quantum theory (1900)
1902	Landsteiner discovers blood groups	First electric typewriter (1901)

Table 1.1 Continued

Date	Microbiological History	Other Historical Events
1903	Wright and others discover antibodies in the blood of immunized animals	First powered aircraft (1903)
1905	Schaudinn and Hoffmann show <i>Treponema pallidum</i> causes syphilis	Einstein's special theory of relativity (1905)
1906	Wassermann develops complement fixation test for syphilis	
1909	Ricketts shows that Rocky Mountain spotted fever is transmitted by ticks and caused by a microbe (<i>Rickettsia rickettsii</i>)	First model T Ford (1908) Peary and Hensen reach North Pole (1909)
1910	Ehrlich develops chemotherapeutic agent for syphilis	Rutherford presents his theory of the atom (1911)
1911	Rous discovers a virus that causes cancer in chickens	Picasso and cubism (1912) World War I begins (1914)
1915–1917	D'Herelle and Twort discover bacterial viruses	Einstein's general theory of relativity (1916) Russian Revolution (1917)
1921	Fleming discovers lysozyme	
1923	First edition of <i>Bergey's Manual</i>	Lindberg's transatlantic flight (1927)
1928	Griffith discovers bacterial transformation	
1929	Fleming discovers penicillin	Stock market crash (1929)
1931	Van Niel shows that photosynthetic bacteria use reduced compounds as electron donors without producing oxygen	
1933	Ruska develops first transmission electron microscope	Hitler becomes chancellor of Germany (1933)
1935	Stanley crystallizes the tobacco mosaic virus Domagk discovers sulfa drugs	
1937	Chatton divides living organisms into procaryotes and eucaryotes	Krebs discovers the citric acid cycle (1937) World War II begins (1939)
1941	Beadle and Tatum, one-gene-one-enzyme hypothesis	
1944	Avery shows that DNA carries information during transformation Waksman discovers streptomycin	The insecticide DDT introduced (1944)
1946	Lederberg and Tatum describe bacterial conjugation	Atomic bombs dropped on Hiroshima and Nagasaki (1945) United Nations formed (1945) First electronic computer (1946)
1949	Enders, Weller, and Robbins grow poliovirus in human tissue cultures	
1950	Lwoff induces lysogenic bacteriophages	Korean War begins (1950)
1952	Hershey and Chase show that bacteriophages inject DNA into host cells Zinder and Lederberg discover generalized transduction	First hydrogen bomb exploded (1952) Stalin dies (1952) First commercial transistorized product (1952)
1953	Phase-contrast microscope developed Medawar discovers immune tolerance Watson and Crick propose the double helix structure for DNA	U.S. Supreme Court rules against segregated schools (1954)
1955	Jacob and Wollman discover the F factor is a plasmid	Montgomery bus boycott (1955)
1959	Jerne and Burnet propose the clonal selection theory	Sputnik launched by Soviet Union (1957)
1959	Yalow develops the radioimmunoassay technique	Birth control pill (1960)
1961	Jacob and Monod propose the operon model of gene regulation	First humans in space (1961)
1961–1966	Nirenberg, Khorana, and others elucidate the genetic code	Cuban missile crisis (1962) Nuclear test ban treaty (1963)
1962	Porter proposes the basic structure for immunoglobulin G First quinolone antimicrobial (nalidixic acid) synthesized	Civil Rights March on Washington (1963) President Kennedy assassinated (1963) Arab-Israeli War (1967) Martin Luther King assassination (1968) Neil Armstrong walks on the moon (1969)
1970	Discovery of restriction endonucleases by Arber and Smith Discovery of reverse transcriptase in retroviruses by Temin and Baltimore	
1973	Ames develops a bacterial assay for the detection of mutagens Cohen, Boyer, Chang, and Helling use plasmid vectors to clone genes in bacteria	Salt I Treaty (1972) Vietnam War ends (1973)
1975	Kohler and Milstein develop technique for the production of monoclonal antibodies	President Nixon resigns because of Watergate cover-up (1974)
1977	Lyme disease discovered Recognition of archaea as a distinct microbial group by Woese and Fox	Panama Canal Treaty (1977)

Table 1.1 Continued

Date	Microbiological History	Other Historical Events
1979	Gilbert and Sanger develop techniques for DNA sequencing Insulin synthesized using recombinant DNA techniques Smallpox declared officially eliminated	Hostages seized in Iran (1978) Three Mile Island disaster (1979) Home computers marketed (1980)
1980	Development of the scanning tunneling microscope	AIDS first recognized (1981)
1982	Recombinant hepatitis B vaccine developed	First artificial heart implanted (1982)
1982–1983	Discovery of catalytic RNA by Cech and Altman	Meter redefined in terms of distance light travels (1983)
1983–1984	The human immunodeficiency virus isolated and identified by Gallo and Montagnier	
	The polymerase chain reaction developed by Mullis	
1986	First vaccine (hepatitis B vaccine) produced by genetic engineering approved for human use	Gorbachev becomes Communist party general secretary (1985) Berlin Wall falls (1989)
1990	First human gene-therapy testing begun	Persian Gulf War with Iraq begins (1990) Soviet Union collapse; Boris Yeltsin comes to power (1991)
1992	First human trials of antisense therapy	
1995	Chickenpox vaccine approved for U.S. use <i>Haemophilus influenzae</i> genome sequenced	
1996	<i>Methanococcus jannaschii</i> genome sequenced Yeast genome sequenced	Water found on the moon (1998)
1997	Discovery of <i>Thiomargarita namibiensis</i> , the largest known bacterium <i>Escherichia coli</i> genome sequenced	
2000	Discovery that <i>Vibrio cholerae</i> has two separate chromosomes	



(a)

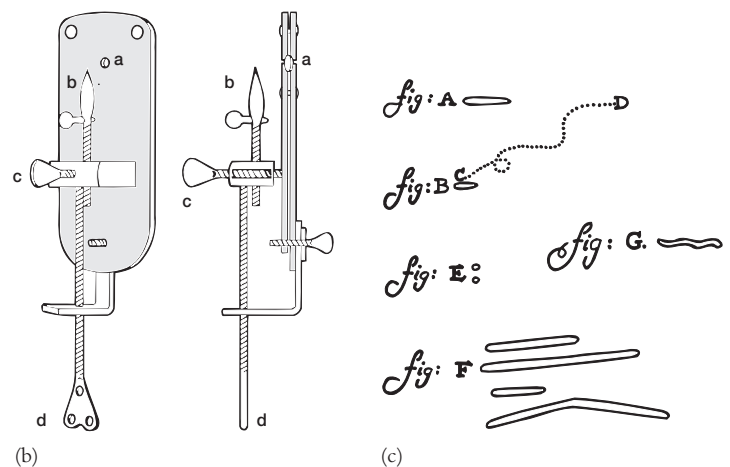


Figure 1.1 **Antony van Leeuwenhoek.** Leeuwenhoek (1632–1723) and his microscopes. (a) Leeuwenhoek holding a microscope. (b) A drawing of one of the microscopes showing the lens, *a*; mounting pin, *b*; and focusing screws, *c* and *d*. (c) Leeuwenhoek's drawings of bacteria from the human mouth. (b) Source: C. E. Dobell, *Antony van Leeuwenhoek and His Little Animals* (1932), Russell and Russell, 1958.

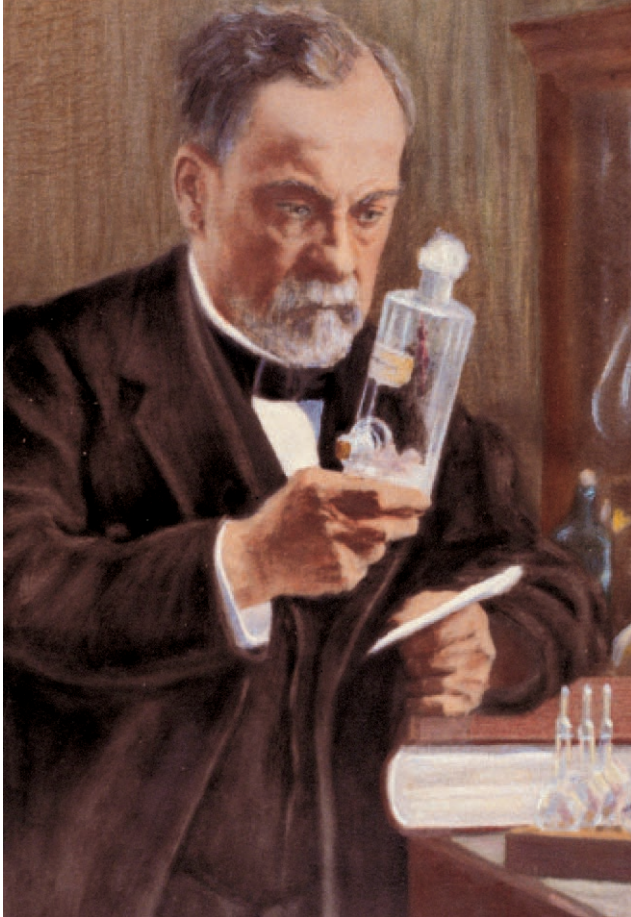


Figure 1.2 Louis Pasteur. Pasteur (1822–1895) working in his laboratory.

Thus the generation of maggots by decaying meat resulted from the presence of fly eggs, and meat did not spontaneously generate maggots as previously believed. Similar experiments by others helped discredit the theory for larger organisms.

Leeuwenhoek's discovery of microorganisms renewed the controversy. Some proposed that microorganisms arose by spontaneous generation even though larger organisms did not. They pointed out that boiled extracts of hay or meat would give rise to microorganisms after sitting for a while. In 1748 the English priest John Needham (1713–1781) reported the results of his experiments on spontaneous generation. Needham boiled mutton broth and then tightly stoppered the flasks. Eventually many of the flasks became cloudy and contained microorganisms. He thought organic matter contained a vital force that could confer the properties of life on nonliving matter. A few years later the Italian priest and naturalist Lazzaro Spallanzani (1729–1799) improved on Needham's experimental design by first sealing glass flasks that contained water and seeds. If the sealed flasks were placed in boiling water for 3/4 of an hour, no growth took place as long as the

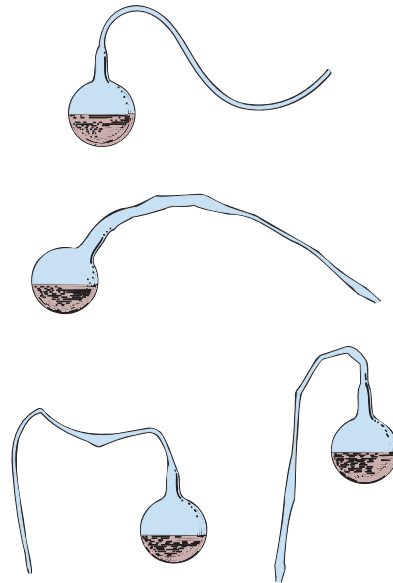


Figure 1.3 The Spontaneous Generation Experiment. Pasteur's swan neck flasks used in his experiments on the spontaneous generation of microorganisms. Source: *Annales Sciences Naturelle, 4th Series, Vol. 16, pp.1–98, Pasteur, L., 1861, "Mémoire sur les Corpuscules Organisés Qui Existent Dans L'Atmosphère: Examen de la Doctrine des Générations Spontanées."*

flasks remained sealed. He proposed that air carried germs to the culture medium, but also commented that the external air might be required for growth of animals already in the medium. The supporters of spontaneous generation maintained that heating the air in sealed flasks destroyed its ability to support life.

Several investigators attempted to counter such arguments. Theodore Schwann (1810–1882) allowed air to enter a flask containing a sterile nutrient solution after the air had passed through a red-hot tube. The flask remained sterile. Subsequently Georg Friedrich Schroder and Theodor von Dusch allowed air to enter a flask of heat-sterilized medium after it had passed through sterile cotton wool. No growth occurred in the medium even though the air had not been heated. Despite these experiments the French naturalist Felix Pouchet claimed in 1859 to have carried out experiments conclusively proving that microbial growth could occur without air contamination. This claim provoked Louis Pasteur (1822–1895) to settle the matter once and for all. Pasteur (**figure 1.2**) first filtered air through cotton and found that objects resembling plant spores had been trapped. If a piece of the cotton was placed in sterile medium after air had been filtered through it, microbial growth appeared. Next he placed nutrient solutions in flasks, heated their necks in a flame, and drew them out into a variety of curves, while keeping the ends of the necks open to the atmosphere (**figure 1.3**). Pasteur then boiled the solutions for a few minutes and allowed them to cool. No growth took place even though the contents of the flasks were exposed to the air. Pasteur pointed out that no growth occurred because dust and germs had been trapped on the

walls of the curved necks. If the necks were broken, growth commenced immediately. Pasteur had not only resolved the controversy by 1861 but also had shown how to keep solutions sterile.

The English physicist John Tyndall (1820–1893) dealt a final blow to spontaneous generation in 1877 by demonstrating that dust did indeed carry germs and that if dust was absent, broth remained sterile even if directly exposed to air. During the course of his studies, Tyndall provided evidence for the existence of exceptionally heat-resistant forms of bacteria. Working independently, the German botanist Ferdinand Cohn (1828–1898) discovered the existence of heat-resistant bacterial endospores (*see chapter 3*).

1. Describe the field of microbiology in terms of the size of its subject material and the nature of its techniques.
2. How did Pasteur and Tyndall finally settle the spontaneous generation controversy?

1.3 The Role of Microorganisms in Disease

The importance of microorganisms in disease was not immediately obvious to people, and it took many years for scientists to establish the connection between microorganisms and illness. Recognition of the role of microorganisms depended greatly upon the development of new techniques for their study. Once it became clear that disease could be caused by microbial infections, microbiologists began to examine the way in which hosts defended themselves against microorganisms and to ask how disease might be prevented. The field of immunology was born.

Recognition of the Relationship between Microorganisms and Disease

Although Fracastoro and a few others had suggested that invisible organisms produced disease, most believed that disease was due to causes such as supernatural forces, poisonous vapors called miasmas, and imbalances between the four humors thought to be present in the body. The idea that an imbalance between the four humors (blood, phlegm, yellow bile [choler], and black bile [melancholy]) led to disease had been widely accepted since the time of the Greek physician Galen (129–199). Support for the germ theory of disease began to accumulate in the early nineteenth century. Agostino Bassi (1773–1856) first showed a microorganism could cause disease when he demonstrated in 1835 that a silkworm disease was due to a fungal infection. He also suggested that many diseases were due to microbial infections. In 1845 M. J. Berkeley proved that the great Potato Blight of Ireland was caused by a fungus. Following his successes with the study of fermentation, Pasteur was asked by the French government to investigate the pébrine disease of silkworms that was disrupting the silk industry. After several years of work, he showed that the disease was due to a protozoan parasite. The disease was controlled by raising caterpillars from eggs produced by healthy moths.



Figure 1.4 Robert Koch. Koch (1843–1910) examining a specimen in his laboratory.

Indirect evidence that microorganisms were agents of human disease came from the work of the English surgeon Joseph Lister (1827–1912) on the prevention of wound infections. Lister impressed with Pasteur's studies on the involvement of microorganisms in fermentation and putrefaction, developed a system of antiseptic surgery designed to prevent microorganisms from entering wounds. Instruments were heat sterilized, and phenol was used on surgical dressings and at times sprayed over the surgical area. The approach was remarkably successful and transformed surgery after Lister published his findings in 1867. It also provided strong indirect evidence for the role of microorganisms in disease because phenol, which killed bacteria, also prevented wound infections.

The first direct demonstration of the role of bacteria in causing disease came from the study of anthrax (*see chapter 39*) by the German physician Robert Koch (1843–1910). Koch (**figure 1.4**) used the criteria proposed by his former teacher, Jacob Henle (1809–1885), to establish the relationship between *Bacillus anthracis* and anthrax, and published his findings in 1876 (**Box 1.1** briefly discusses the scientific method). Koch injected healthy mice with material from diseased animals, and the mice became ill. After transferring anthrax by inoculation through a series of 20 mice, he incubated a piece of spleen containing the anthrax bacillus in beef serum. The bacilli grew, reproduced, and produced spores. When the isolated bacilli or spores were injected into mice, anthrax developed. His criteria for proving the causal relationship between a microorganism and a specific disease are known as **Koch's postulates** and can be summarized as follows:

1. The microorganism must be present in every case of the disease but absent from healthy organisms.

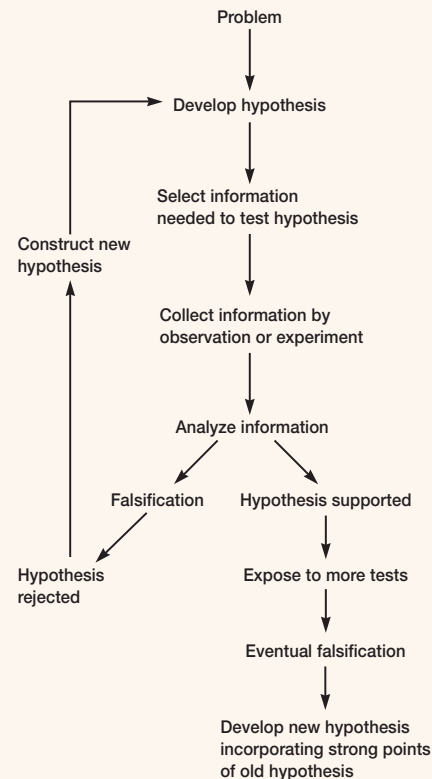
Box 1.1

The Scientific Method

Although biologists employ a variety of approaches in conducting research, microbiologists and other experimentally oriented biologists often use the general approach known as the scientific method. They first gather observations of the process to be studied and then develop a tentative **hypothesis**—an educated guess—to explain the observations (see **Box figure**). This step often is inductive and creative because there is no detailed, automatic technique for generating hypotheses. Next they decide what information is required to test the hypothesis and collect this information through observation or carefully designed experiments. After the information has been collected, they decide whether the hypothesis has been supported or falsified. If it has failed to pass the test, the hypothesis is rejected, and a new explanation or hypothesis is constructed. If the hypothesis passes the test, it is subjected to more severe testing. The procedure often is made more efficient by constructing and testing alternative hypotheses and then refining the hypothesis that survives testing. This general approach is often called the hypothetico-deductive method. One deduces predictions from the currently accepted hypothesis and tests them. In deduction the conclusion about specific cases follows logically from a general premise (“if . . . , then . . .” reasoning). Induction is the opposite. A general conclusion is reached after considering many specific examples. Both types of reasoning are used by scientists.

When carrying out an experiment, it is essential to use a control group as well as an experimental group. The control group is treated precisely the same as the experimental group except that the experimental manipulation is not performed on it. In this way one can be sure that any changes in the experimental group are due to the experimental manipulation rather than to some other factor not taken into account.

If a hypothesis continues to survive testing, it may be accepted as a valid theory. A **theory** is a set of propositions and concepts that provides a reliable, systematic, and rigorous account of an aspect of nature. It is important to note that hypotheses and theories are never absolutely proven. Scientists simply gain more and more confidence in their accuracy as they continue to survive testing, fit with new observations and experiments, and satisfactorily explain the observed phenomena.



The Hypothetico-Deductive Method. This approach is most often used in scientific research.

2. The suspected microorganism must be isolated and grown in a pure culture.
3. The same disease must result when the isolated microorganism is inoculated into a healthy host.
4. The same microorganism must be isolated again from the diseased host.

Although Koch used the general approach described in the postulates during his anthrax studies, he did not outline them fully until his 1884 publication on the cause of tuberculosis (**Box 1.2**).

Koch's proof that *Bacillus anthracis* caused anthrax was independently confirmed by Pasteur and his coworkers. They discovered that after burial of dead animals, anthrax spores survived and were brought to the surface by earthworms. Healthy animals then ingested the spores and became ill.

The Development of Techniques for Studying Microbial Pathogens

During Koch's studies on bacterial diseases, it became necessary to isolate suspected bacterial pathogens. At first he cultured bacteria on the sterile surfaces of cut, boiled potatoes. This was unsatisfactory because bacteria would not always grow well on potatoes. He then tried to solidify regular liquid media by adding gelatin. Separate bacterial colonies developed after the surface had been streaked with a bacterial sample. The sample could also be mixed with liquefied gelatin medium. When the gelatin medium hardened, individual bacteria produced separate colonies. Despite its advantages gelatin was not an ideal solidifying agent because it was digested by many bacteria and melted when the temperature rose above 28°C. A better alternative was provided by Fannie

Box 1.2

Molecular Koch's Postulates

Although the criteria that Koch developed for proving a causal relationship between a microorganism and a specific disease have been of immense importance in medical microbiology, it is not always possible to apply them in studying human diseases. For example, some pathogens cannot be grown in pure culture outside the host; because other pathogens grow only in humans, their study would require experimentation on people. The identification, isolation, and cloning of genes responsible for pathogen virulence (*see p. 794*) have made possible a new molecular form of Koch's postulates that resolves some of these difficulties. The emphasis is on the virulence genes present in the infectious agent rather than on the agent itself. The molecular postulates can be briefly summarized as follows:

1. The virulence trait under study should be associated much more with pathogenic strains of the species than with nonpathogenic strains.

2. Inactivation of the gene or genes associated with the suspected virulence trait should substantially decrease pathogenicity.
3. Replacement of the mutated gene with the normal wild-type gene should fully restore pathogenicity.
4. The gene should be expressed at some point during the infection and disease process.
5. Antibodies or immune system cells directed against the gene products should protect the host.

The molecular approach cannot always be applied because of problems such as the lack of an appropriate animal system. It also is difficult to employ the molecular postulates when the pathogen is not well characterized genetically.

Eilshemius Hesse, the wife of Walther Hesse, one of Koch's assistants (**figure 1.5**). She suggested the use of agar as a solidifying agent—she had been using it successfully to make jellies for some time. Agar was not attacked by most bacteria and did not melt until reaching a temperature of 100°C. One of Koch's assistants, Richard Petri, developed the petri dish (plate), a container for solid culture media. These developments made possible the isolation of pure cultures that contained only one type of bacterium, and directly stimulated progress in all areas of bacteriology. *Isolation of bacteria and pure culture techniques* (pp. 106–10).

Koch also developed media suitable for growing bacteria isolated from the body. Because of their similarity to body fluids, meat extracts and protein digests were used as nutrient sources. The result was the development of nutrient broth and nutrient agar, media that are still in wide use today.

By 1882 Koch had used these techniques to isolate the bacillus that caused tuberculosis. There followed a golden age of about 30 to 40 years in which most of the major bacterial pathogens were isolated (table 1.1).

The discovery of viruses and their role in disease was made possible when Charles Chamberland (1851–1908), one of Pasteur's associates, constructed a porcelain bacterial filter in 1884. The first viral pathogen to be studied was the tobacco mosaic disease virus (*see chapter 16*). *The development of virology* (pp. 362–63).

Immunological Studies

In this period progress also was made in determining how animals resisted disease and in developing techniques for protecting humans and livestock against pathogens. During studies on chicken cholera, Pasteur and Roux discovered that incubating their cultures for long intervals between transfers would attenuate the bacteria, which meant they had lost their ability to cause



Figure 1.5 Fannie Eilshemius (1850–1934) and Walther Hesse (1846–1911). Fannie Hesse first proposed using agar in culture media.

the disease. If the chickens were injected with these attenuated cultures, they remained healthy but developed the ability to resist the disease. He called the attenuated culture a vaccine [Latin *vacca*, cow] in honor of Edward Jenner because, many years earlier, Jenner had used vaccination with material from cowpox lesions to protect people against smallpox (*see section 16.1*). Shortly after this, Pasteur and Chamberland developed an attenuated anthrax vaccine in two ways: by treating cultures with potassium bichromate and by incubating the bacteria at 42 to 43°C. *Vaccines and immunizations* (pp. 764–68).

Pasteur next prepared rabies vaccine by a different approach. The pathogen was attenuated by growing it in an abnormal host, the rabbit. After infected rabbits had died, their brains and spinal cords were removed and dried. During the course of these studies, Joseph Meister, a nine-year-old boy who had been bitten by a rabid dog, was brought to Pasteur. Since the boy's death was certain in the absence of treatment, Pasteur agreed to try vaccination. Joseph was injected 13 times over the next 10 days with increasingly virulent preparations of the attenuated virus. He survived.

In gratitude for Pasteur's development of vaccines, people from around the world contributed to the construction of the Pasteur Institute in Paris, France. One of the initial tasks of the Institute was vaccine production.

After the discovery that the diphtheria bacillus produced a toxin, Emil von Behring (1854–1917) and Shibasaburo Kitasato (1852–1931) injected inactivated toxin into rabbits, inducing them to produce an antitoxin, a substance in the blood that would inactivate the toxin and protect against the disease. A tetanus antitoxin was then prepared and both antitoxins were used in the treatment of people.

The antitoxin work provided evidence that immunity could result from soluble substances in the blood, now known to be antibodies (humoral immunity). It became clear that blood cells were also important in immunity (cellular immunity) when Elie Metchnikoff (1845–1916) discovered that some blood leukocytes could engulf disease-causing bacteria (**figure 1.6**). He called these cells phagocytes and the process phagocytosis [Greek *phagein*, eating].



Figure 1.6 Elie Metchnikoff. Metchnikoff (1845–1916) shown here at work in his laboratory.

1. Discuss the contributions of Lister, Pasteur, and Koch to the germ theory of disease and to the treatment or prevention of diseases.
2. What other contributions did Koch make to microbiology?
3. Describe Koch's postulates. What are the molecular Koch's postulates and why are they important?
4. How did von Behring and Metchnikoff contribute to the development of immunology?

1.4 Industrial Microbiology and Microbial Ecology

Although Theodore Schwann and others had proposed in 1837 that yeast cells were responsible for the conversion of sugars to alcohol, a process they called alcoholic fermentation, the leading chemists of the time believed microorganisms were not involved. They were convinced that fermentation was due to a chemical instability that degraded the sugars to alcohol. Pasteur did not agree. It appears that early in his career Pasteur became interested in fermentation because of his research on the stereochemistry of molecules. He believed that fermentations were carried out by living organisms and produced asymmetric products such as amyl alcohol that had optical activity. There was an intimate connection between molecular asymmetry, optical activity, and life. Then in 1856 M. Bigo, an industrialist in Lille, France, where Pasteur worked, requested Pasteur's assistance.

His business produced ethanol from the fermentation of beet sugars, and the alcohol yields had recently declined and the product had become sour. Pasteur discovered that the fermentation was failing because the yeast normally responsible for alcohol formation had been replaced by microorganisms producing lactic acid rather than ethanol. In solving this practical problem, Pasteur demonstrated that all fermentations were due to the activities of specific yeasts and bacteria, and he published several papers on fermentation between 1857 and 1860. His success led to a study of wine diseases and the development of pasteurization (*see chapter 7*) to preserve wine during storage. Pasteur's studies on fermentation continued for almost 20 years. One of his most important discoveries was that some fermentative microorganisms were anaerobic and could live only in the absence of oxygen, whereas others were able to live either aerobically or anaerobically. [Fermentation \(pp. 179–81\); The effect of oxygen on microorganisms \(pp. 127–29\).](#)

A few of the early microbiologists chose to investigate the ecological role of microorganisms. In particular they studied microbial involvement in the carbon, nitrogen, and sulfur cycles taking place in soil and aquatic habitats. Two of the pioneers in this endeavor were Sergei N. Winogradsky (1856–1953) and Martinus W. Beijerinck (1851–1931). [Biogeochemical cycles \(pp. 611–18\).](#)

The Russian microbiologist Sergei N. Winogradsky made many contributions to soil microbiology. He discovered that soil bacteria could oxidize iron, sulfur, and ammonia to obtain energy, and that many bacteria could incorporate CO₂ into organic matter much like photosynthetic organisms do. Winogradsky also isolated anaerobic nitrogen-fixing soil bacteria and studied the decomposition of cellulose.

Martinus W. Beijerinck was one of the great general microbiologists who made fundamental contributions to microbial ecology and many other fields. He isolated the aerobic nitrogen-fixing bacterium *Azotobacter*; a root nodule bacterium also capable of fixing nitrogen (later named *Rhizobium*); and sulfate-reducing bacteria. Beijerinck and Winogradsky developed the enrichment-culture technique and the use of selective media (see chapter 5), which have been of such great importance in microbiology.

1. Briefly describe the work of Pasteur on microbial fermentations.
2. How did Winogradsky and Beijerinck contribute to the study of microbial ecology?

1.5 Members of the Microbial World

Although the kingdoms of organisms and the differences between procaryotic and eucaryotic cells are discussed in much more detail later, a brief introduction to the organisms a microbiologist studies is given here. [Comparison of procaryotic and eucaryotic cells \(pp. 91–92\).](#)

Two fundamentally different types of cells exist. **Procaryotic cells** [Greek *pro*, before, and *karyon*, nut or kernel; organism with a primordial nucleus] have a much simpler morphology than eucaryotic cells and lack a true membrane-delimited nucleus. All bacteria are procaryotic. In contrast, **eucaryotic cells** [Greek *eu*, true, and *karyon*, nut or kernel] have a membrane-enclosed nucleus; they are more complex morphologically and are usually larger than procaryotes. Algae, fungi, protozoa, higher plants, and animals are eucaryotic. Procaryotic and eucaryotic cells differ in many other ways as well (see chapter 4).

The early description of organisms as either plants or animals clearly is too simplified, and for many years biologists have divided organisms into five kingdoms: the *Monera*, *Protista*, *Fungi*, *Animalia*, and *Plantae* (see chapter 19). Microbiologists study primarily members of the first three kingdoms. Although they are not included in the five kingdoms, viruses are also studied by microbiologists. [Fungi \(chapter 25\); Algae \(chapter 26\); Protozoa \(chapter 27\); Introduction to the viruses \(chapters 16–18\)](#)

In the last few decades great progress has been made in three areas that profoundly affect microbial classification. First, much has been learned about the detailed structure of microbial cells from the use of electron microscopy. Second, microbiologists have determined the biochemical and physiological characteristics of many different microorganisms. Third, the sequences of nucleic acids and proteins from a wide variety of

organisms have been compared. It is now clear that there are two quite different groups of procaryotic organisms: Bacteria and Archaea. Furthermore, the protists are so diverse that it may be necessary to divide the kingdom *Protista* into three or more kingdoms. Thus many taxonomists have concluded that the five-kingdom system is too simple and have proposed a variety of alternatives (see section 19.7). The differences between Bacteria, Archaea, and the eucaryotes seem so great that many microbiologists have proposed that organisms should be divided among three domains: Bacteria (the true bacteria or eubacteria), Archaea¹, and Eucarya (all eucaryotic organisms). This system, which we shall use here, and the results leading to it are discussed in chapter 19.

1. Describe and contrast procaryotic and eucaryotic cells.
2. Briefly describe the five-kingdom system and give the major characteristics of each kingdom.

1.6 The Scope and Relevance of Microbiology

As the scientist-writer Steven Jay Gould emphasized, we live in the Age of Bacteria. They were the first living organisms on our planet, live virtually everywhere life is possible, are more numerous than any other kind of organism, and probably constitute the largest component of the earth's biomass. The whole ecosystem depends on their activities, and they influence human society in countless ways. Thus modern microbiology is a large discipline with many different specialties; it has a great impact on fields such as medicine, agricultural and food sciences, ecology, genetics, biochemistry, and molecular biology.

For example, microbiology has been a major contributor to the rise of molecular biology, the branch of biology dealing with the physical and chemical aspects of living matter and its function. Microbiologists have been deeply involved in studies on the genetic code and the mechanisms of DNA, RNA, and protein synthesis. Microorganisms were used in many of the early studies on the regulation of gene expression and the control of enzyme activity (see chapters 8 and 12). In the 1970s new discoveries in microbiology led to the development of recombinant DNA technology and genetic engineering. [The mechanisms of DNA, RNA, and protein synthesis \(chapters 11 and 12\); Recombinant DNA and genetic engineering \(chapter 14\)](#)

One indication of the importance of microbiology in the twentieth century is the Nobel Prize given for work in physiology or medicine. About 1/3 of these have been awarded to scientists working on microbiological problems (see inside front cover).

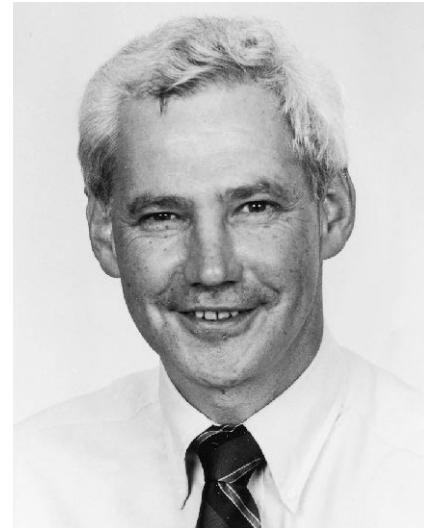
¹Although this will be discussed further in chapter 19, it should be noted here that several names have been used for the Archaea. The two most important are archaeobacteria and eubacteria. In this text, we shall use only the name Archaea for sake of clarity and consistency.



(a)



(b)



(c)



(d)



(e)



(f)

Figure 1.7 Some Well-Known Modern Microbiologists. This figure depicts a few microbiologists who have made significant contributions in different areas of microbiology. (a) Rita R. Colwell has studied the genetics and ecology of marine bacteria such as *Vibrio cholerae* and helped establish the field of marine biotechnology. (b) R. G. E. Murray has contributed greatly to the understanding of bacterial cell envelopes and bacterial taxonomy. (c) Stanley Falkow has advanced our understanding of how bacterial pathogens cause disease. (d) Martha Howe has made fundamental contributions to our knowledge of the bacteriophage Mu. (e) Frederick C. Neidhardt has contributed to microbiology through his work on the regulation of *E. coli* physiology and metabolism, and by coauthoring advanced textbooks. (f) Jean E. Brechley has studied the regulation of glutamate and glutamine metabolism, helped found the Pennsylvania State University Biotechnology Institute, and is now finding biotechnological uses for psychrophilic (cold-loving) microorganisms.

Microbiology has both basic and applied aspects. Many microbiologists are interested primarily in the biology of the microorganisms themselves (**figure 1.7**). They may focus on a specific group of microorganisms and be called virologists (viruses), bacteriologists (bacteria), phycologists or algologists (algae), mycologists (fungi), or protozoologists (protozoa). Others are interested in microbial

morphology or particular functional processes and work in fields such as microbial cytology, microbial physiology, microbial ecology, microbial genetics and molecular biology, and microbial taxonomy. Of course a person can be thought of in both ways (e.g., as a bacteriologist who works on taxonomic problems). Many microbiologists have a more applied orientation and work on practical

problems in fields such as medical microbiology, food and dairy microbiology, and public health microbiology (basic research is also conducted in these fields). Because the various fields of microbiology are interrelated, an applied microbiologist must be familiar with basic microbiology. For example, a medical microbiologist must have a good understanding of microbial taxonomy, genetics, immunology, and physiology to identify and properly respond to the pathogen of concern.

What are some of the current occupations of professional microbiologists? One of the most active and important is medical microbiology, which deals with the diseases of humans and animals. Medical microbiologists identify the agent causing an infectious disease and plan measures to eliminate it. Frequently they are involved in tracking down new, unidentified pathogens such as the agent that causes variant creutzfeldt-Jacob disease, the hantavirus, and the virus responsible for AIDS. These microbiologists also study the ways in which microorganisms cause disease. [Legionnaires' disease \(pp. 901–2\); Hantavirus pulmonary syndrome \(p. 877\); AIDS \(pp. 878–84\)](#)

Public health microbiology is closely related to medical microbiology. Public health microbiologists try to control the spread of communicable diseases. They often monitor community food establishments and water supplies in an attempt to keep them safe and free from infectious disease agents.

Immunology is concerned with how the immune system protects the body from pathogens and the response of infectious agents. It is one of the fastest growing areas in science; for example, techniques for the production and use of monoclonal antibodies have developed extremely rapidly. Immunology also deals with practical health problems such as the nature and treatment of allergies and autoimmune diseases like rheumatoid arthritis. [Monoclonal antibodies and their uses \(section 32.3 and Box 36.2\)](#)

Many important areas of microbiology do not deal directly with human health and disease but certainly contribute to human welfare. Agricultural microbiology is concerned with the impact of microorganisms on agriculture. Agricultural microbiologists try to combat plant diseases that attack important food crops, work on methods to increase soil fertility and crop yields, and study the role of microorganisms living in the digestive tracts of ruminants such as cattle. Currently there is great interest in using bacterial and viral insect pathogens as substitutes for chemical pesticides.

The field of microbial ecology is concerned with the relationships between microorganisms and their living and nonliving habitats. Microbial ecologists study the contributions of microorganisms to the carbon, nitrogen, and sulfur cycles in soil and in freshwater. The study of pollution effects on microorganisms also is important because of the impact these organisms have on the environment. Microbial ecologists are employing microorganisms in bioremediation to reduce pollution effects.

Scientists working in food and dairy microbiology try to prevent microbial spoilage of food and the transmission of food-borne diseases such as botulism and salmonellosis ([see chapter 39](#)). They also use microorganisms to make foods such as cheeses, yogurts, pickles, and beer. In the future microorganisms themselves may become a more important nutrient source for livestock and humans.

In industrial microbiology microorganisms are used to make products such as antibiotics, vaccines, steroids, alcohols and other solvents, vitamins, amino acids, and enzymes. Microorganisms can even leach valuable minerals from low-grade ores.

Research on the biology of microorganisms occupies the time of many microbiologists and also has practical applications. Those working in microbial physiology and biochemistry study the synthesis of antibiotics and toxins, microbial energy production, the ways in which microorganisms survive harsh environmental conditions, microbial nitrogen fixation, the effects of chemical and physical agents on microbial growth and survival, and many other topics.

Microbial genetics and molecular biology focus on the nature of genetic information and how it regulates the development and function of cells and organisms. The use of microorganisms has been very helpful in understanding gene function. Microbial geneticists play an important role in applied microbiology by producing new microbial strains that are more efficient in synthesizing useful products. Genetic techniques are used to test substances for their ability to cause cancer. More recently the field of genetic engineering ([see chapter 14](#)) has arisen from work in microbial genetics and molecular biology and will contribute substantially to microbiology, biology as a whole, and medicine. Engineered microorganisms are used to make hormones, antibiotics, vaccines, and other products ([see chapter 42](#)). New genes can be inserted into plants and animals; for example, it may be possible to give corn and wheat nitrogen-fixation genes so they will not require nitrogen fertilizers.

1.7 The Future of Microbiology

As the preceding sections have shown, microbiology has had a profound influence on society. What of the future? Science writer Bernard Dixon is very optimistic about microbiology's future for two reasons. First, microbiology has a clearer mission than do many other scientific disciplines. Second, it is confident of its value because of its practical significance. Dixon notes that microbiology is required both to face the threat of new and reemerging human infectious diseases and to develop industrial technologies that are more efficient and environmentally friendly.

What are some of the most promising areas for future microbiological research and their potential practical impacts? What kinds of challenges do microbiologists face? The following brief list should give some idea of what the future may hold:

1. New infectious diseases are continually arising and old diseases are once again becoming widespread and destructive. AIDS, hemorrhagic fevers, and tuberculosis are excellent examples of new and reemerging infectious diseases. Microbiologists will have to respond to these threats, many of them presently unknown.
2. Microbiologists must find ways to stop the spread of established infectious diseases. Increases in antibiotic resistance will be a continuing problem, particularly the spread of multiple drug resistance that can render a pathogen impervious to current medical treatment.

Microbiologists have to create new drugs and find ways to slow or prevent the spread of drug resistance. New vaccines must be developed to protect against diseases such as AIDS. It will be necessary to use techniques in molecular biology and recombinant DNA technology to solve these problems.

3. Research is needed on the association between infectious agents and chronic diseases such as autoimmune and cardiovascular diseases. It may be that some of these chronic afflictions partly result from infections.
4. We are only now beginning to understand how pathogens interact with host cells and the ways in which diseases arise. There also is much to learn about how the host resists pathogen invasions.
5. Microorganisms are increasingly important in industry and environmental control, and we must learn how to use them in a variety of new ways. For example, microorganisms can (a) serve as sources of high-quality food and other practical products such as enzymes for industrial applications, (b) degrade pollutants and toxic wastes, and (c) be used as vectors to treat diseases and enhance agricultural productivity. There also is a continuing need to protect food and crops from microbial damage.
6. Microbial diversity is another area requiring considerable research. Indeed, it is estimated that less than 1% of the earth's microbial population has been cultured. We must develop new isolation techniques and an adequate classification of microorganisms, one which includes those microbes that cannot be cultivated in the laboratory. Much work needs to be done on microorganisms living in extreme environments. The discovery of new microorganisms may well lead to further advances in industrial processes and enhanced environmental control.
7. Microbial communities often live in biofilms, and these biofilms are of profound importance in both medicine and microbial ecology. Research on biofilms is in its infancy; it will be many years before we more fully understand their nature and are able to use our knowledge in practical ways. In general, microbe-microbe interactions have not yet been extensively explored.
8. The genomes of many microorganisms already have been sequenced, and many more will be determined in the

coming years. These sequences are ideal for learning how the genome is related to cell structure and what the minimum assortment of genes necessary for life is. Analysis of the genome and its activity will require continuing advances in the field of bioinformatics and the use of computers to investigate biological problems.

9. Further research on unusual microorganisms and microbial ecology will lead to a better understanding of the interactions between microorganisms and the inanimate world. Among other things, this understanding should enable us to more effectively control pollution. Similarly, it has become clear that microorganisms are essential partners with higher organisms in symbiotic relationships. Greater knowledge of symbiotic relationships can help improve our appreciation of the living world. It also will lead to improvements in the health of plants, livestock, and humans.
10. Because of their relative simplicity, microorganisms are excellent subjects for the study of a variety of fundamental questions in biology. For example, how do complex cellular structures develop and how do cells communicate with one another and respond to the environment?
11. Finally, microbiologists will be challenged to carefully assess the implications of new discoveries and technological developments. They will need to communicate a balanced view of both the positive and negative long-term impacts of these events on society.

The future of microbiology is bright. The microbiologist René Dubos has summarized well the excitement and promise of microbiology:

How extraordinary that, all over the world, microbiologists are now involved in activities as different as the study of gene structure, the control of disease, and the industrial processes based on the phenomenal ability of microorganisms to decompose and synthesize complex organic molecules. Microbiology is one of the most rewarding of professions because it gives its practitioners the opportunity to be in contact with all the other natural sciences and thus to contribute in many different ways to the betterment of human life.

Summary

1. Microbiology may be defined in terms of the size of the organisms studied and the techniques employed.
2. Antony van Leeuwenhoek was the first person to describe microorganisms.
3. Experiments by Redi and others disproved the theory of spontaneous generation in regard to larger organisms.
4. The spontaneous generation of microorganisms was disproved by Spallanzani, Pasteur, Tyndall, and others.
5. Support for the germ theory of disease came from the work of Bassi, Pasteur, Koch, and others. Lister provided indirect evidence with his development of antiseptic surgery.
6. Koch's postulates and molecular Koch's postulates are used to prove a direct relationship between a suspected pathogen and a disease.
7. Koch developed the techniques required to grow bacteria on solid media and to isolate pure cultures of pathogens.
8. Vaccines against anthrax and rabies were made by Pasteur; von Behring and Kitasato prepared antitoxins for diphtheria and tetanus.
9. Metchnikoff discovered some blood leukocytes could phagocytize and destroy bacterial pathogens.
10. Pasteur showed that fermentations were caused by microorganisms and that some microorganisms could live in the absence of oxygen.

11. The role of microorganisms in carbon, nitrogen, and sulfur cycles was first studied by Winogradsky and Beijerinck.
12. Prokaryotic cells differ from eukaryotic cells in lacking a membrane-delimited nucleus, and in other ways as well.
13. The Archaea are so different that many microbiologists divide organisms into three domains: *Bacteria*, *Archaea*, and *Eucarya*.
14. In the twentieth century microbiology has contributed greatly to the fields of biochemistry and genetics. It also has helped stimulate the rise of molecular biology.
15. There is a wide variety of fields in microbiology, and many have a great impact on society. These include the more applied disciplines such as medical, public health, industrial, food, and dairy microbiology. Microbial ecology, physiology, biochemistry, and genetics are examples of basic microbiological research fields.
16. Microbiologists will be faced with many exciting and important future challenges such as finding new ways to combat disease, reduce pollution, and feed the world's population.

Key Terms

eukaryotic cell 11
hypothesis 8
Koch's postulates 7

microbiology 2
microorganism 2
prokaryotic cell 11

spontaneous generation 2
theory 8

Questions for Thought and Review

1. Why was the belief in spontaneous generation an obstacle to the development of microbiology as a scientific discipline?
2. Describe the major contributions of the following people to the development of microbiology: Leeuwenhoek, Spallanzani, Fracastoro, Pasteur, Tyndall, Cohn, Bassi, Lister, Koch, Chamberland, von Behring, Metchnikoff, Winogradsky, and Beijerinck.
3. Would microbiology have developed more slowly if Fannie Hesse had not suggested the use of agar? Give your reasoning. What is a pure culture?
4. Why do you think viruses are not included in the five-kingdom or three domain systems?
5. Why are microorganisms so useful to biologists as experimental models?
6. What do you think were the most important discoveries in the development of microbiology? Why?
7. List all the activities or businesses you can think of in your community that are directly dependent on microbiology.
8. Describe in your own words the scientific method. How does a theory differ from a hypothesis? Why is it important to have a control group?
9. What do you think are the five most important research areas to pursue in microbiology? Give reasons for your choices.

Critical Thinking Questions

1. Consider the impact of microbes on the course of world history. History is full of examples of instances or circumstances under which one group of people lost a struggle against another. In fact, when examined more closely, the "losers" often had the misfortune of being exposed to, more susceptible to, or unable to cope with an infectious agent. Thus, weakened in physical strength or demoralized by the course of a devastating disease, they were easily overcome by human "conquerors."
 - a. Choose an example of a battle or other human activity such as exploration of new territory and determine the impact of microorganisms, either indigenous or transported to the region, on that activity.
 - b. Discuss the effect that the microbe(s) had on the outcome in your example.
 - c. Suggest whether the advent of antibiotics, food storage or preparation technology, or sterilization technology would have made a difference in the outcome.
2. Vaccinations against various childhood diseases have contributed to the entry of women, particularly mothers, into the full-time workplace.
 - a. Is this statement supported by data—comparing availability and extent of vaccination with employment statistics in different places or at different times?
 - b. Before vaccinations for measles, mumps, and chickenpox, what was the incubation time and duration of these childhood diseases? What impact would such diseases have on mothers with several elementary schoolchildren at home if they had full-time jobs and lacked substantial child care support?
 - c. What would be the consequence if an entire generation of children (or a group of children in one country) were not vaccinated against any diseases? What do you predict would happen if these children went to college and lived in a dormitory in close proximity with others who had received all of the recommended childhood vaccines?

Additional Reading

General

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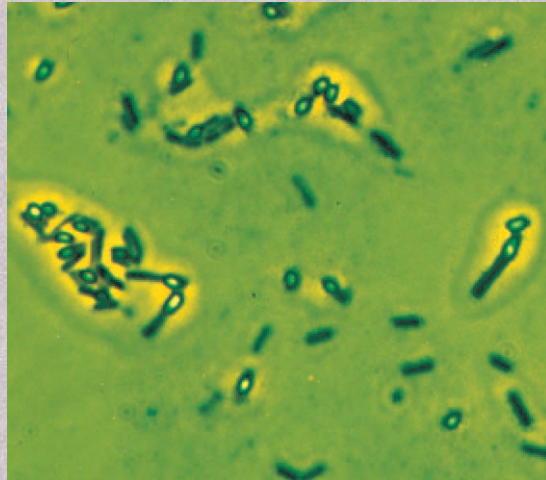
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CHAPTER 2

The Study of Microbial Structure: Microscopy and Specimen Preparation



Clostridium botulinum is a rod-shaped bacterium that forms endospores and releases botulinum toxin, the cause of botulism food poisoning. In this phase-contrast micrograph, the endospores are the bright, oval objects located at the ends of the rods; some endospores have been released from the cells that formed them.

Outline

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Concepts

1. Light microscopes use glass lenses to bend and focus light rays and produce enlarged images of small objects. The resolution of a light microscope is determined by the numerical aperture of its lens system and by the wavelength of the light it employs; maximum resolution is about 0.2 μm .
2. The most common types of light microscopes are the bright-field, dark-field, phase-contrast, and fluorescence microscopes. Each yields a distinctive image and may be used to observe different aspects of microbial morphology.
3. Because most microorganisms are colorless and therefore not easily seen in the bright-field microscope, they are usually fixed and stained before observation. Either simple or differential staining can be used to enhance contrast. Specific bacterial structures such as capsules, endospores, and flagella also can be selectively stained.
4. The transmission electron microscope achieves great resolution (about 0.5 nm) by using electron beams of very short wavelength rather than visible light. Although one can prepare microorganisms for observation in other ways, one normally views thin sections of plastic-embedded specimens treated with heavy metals to improve contrast.
5. External features can be observed in great detail with the scanning electron microscope, which generates an image by scanning a fine electron beam over the surface of specimens rather than projecting electrons through them.
6. New forms of microscopy are improving our ability to observe microorganisms and molecules. Two examples are the confocal scanning laser microscope and the scanning probe microscope.

There are more animals living in the scum on the teeth in a man's mouth than there are men in a whole kingdom.

—Antony van Leeuwenhoek

Microbiology usually is concerned with organisms so small they cannot be seen distinctly with the unaided eye. Because of the nature of this discipline, the microscope is of crucial importance. Thus it is important to understand how the microscope works and the way in which specimens are prepared for examination.

The chapter begins with a detailed treatment of the standard bright-field microscope and then describes other common types of light microscopes. Next preparation and staining of specimens for examination with the light microscope are discussed. This is followed by a description of transmission and scanning electron microscopes, both of which are used extensively in current microbiological research. The chapter closes with a brief introduction to two newer forms of microscopy: scanning probe microscopy and confocal microscopy.

2.1 Lenses and the Bending of Light

To understand how a light microscope operates, one must know something about the way in which lenses bend and focus light to form images. When a ray of light passes from one medium to another, **refraction** occurs—that is, the ray is bent at the interface. The **refractive index** is a measure of how greatly a substance

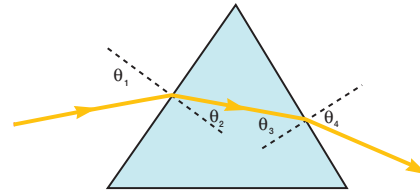


Figure 2.1 The Bending of Light by a Prism. Normals (lines perpendicular to the surface of the prism) are indicated by dashed lines. As light enters the glass, it is bent toward the first normal (angle θ_2 is less than θ_1). When light leaves the glass and returns to air, it is bent away from the second normal (θ_4 is greater than θ_3). As a result the prism bends light passing through it.

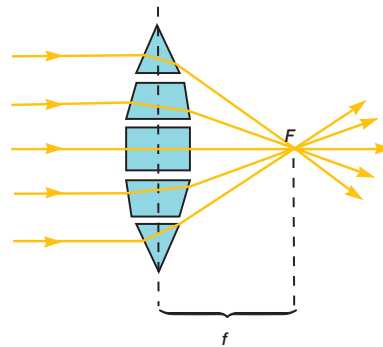


Figure 2.2 Lens Function. A lens functions somewhat like a collection of prisms. Light rays from a distant source are focused at the focal point F . The focal point lies a distance f , the focal length, from the lens center.

slows the velocity of light, and the direction and magnitude of bending is determined by the refractive indexes of the two media forming the interface. When light passes from air into glass, a medium with a greater refractive index, it is slowed and bent toward the normal, a line perpendicular to the surface (**figure 2.1**). As light leaves glass and returns to air, a medium with a lower refractive index, it accelerates and is bent away from the normal. Thus a prism bends light because glass has a different refractive index from air, and the light strikes its surface at an angle.

Lenses act like a collection of prisms operating as a unit. When the light source is distant so that parallel rays of light strike the lens, a convex lens will focus these rays at a specific point, the **focal point** (F in **figure 2.2**). The distance between the center of the lens and the focal point is called the **focal length** (f in **figure 2.2**).

Our eyes cannot focus on objects nearer than about 25 cm or 10 inches (**table 2.1**). This limitation may be overcome by using a convex lens as a simple magnifier (or microscope) and holding it close to an object. A magnifying glass provides a clear image at much closer range, and the object appears larger. Lens strength is related to focal length; a lens with a short focal length will magnify an object more than a weaker lens having a longer focal length.

Table 2.1 Common Units of Measurement

Unit	Abbreviation	Value
1 centimeter	cm	10^{-2} meter or 0.394 inches
1 millimeter	mm	10^{-3} meter
1 micrometer	μm	10^{-6} meter
1 nanometer	nm	10^{-9} meter
1 Angstrom	$\text{\AA}$	10^{-10} meter

1. Define refraction, refractive index, focal point, and focal length.
2. Describe the path of a light ray through a prism or lens.
3. How is lens strength related to focal length?

2.2 The Light Microscope

Microbiologists currently employ a variety of light microscopes in their work; bright-field, dark-field, phase-contrast, and fluorescence microscopes are most commonly used. Modern microscopes are all compound microscopes. That is, the magnified image formed by the objective lens is further enlarged by one or more additional lenses.

The Bright-Field Microscope

The ordinary microscope is called a **bright-field microscope** because it forms a dark image against a brighter background. The microscope consists of a sturdy metal body or stand composed of a base and an arm to which the remaining parts are attached (**figure 2.3**). A light source, either a mirror or an electric illuminator, is located in the base. Two focusing knobs, the fine and coarse adjustment knobs, are located on the arm and can move either the stage or the nosepiece to focus the image.

The stage is positioned about halfway up the arm and holds microscope slides by either simple slide clips or a mechanical stage clip. A mechanical stage allows the operator to move a slide around smoothly during viewing by use of stage control knobs. The **substage condenser** is mounted within or beneath the stage and focuses a cone of light on the slide. Its position often is fixed in simpler microscopes but can be adjusted vertically in more advanced models.

The curved upper part of the arm holds the body assembly, to which a nosepiece and one or more **eyepieces** or **oculars** are attached. More advanced microscopes have eyepieces for both eyes and are called binocular microscopes. The body assembly itself contains a series of mirrors and prisms so that the barrel holding the eyepiece may be tilted for ease in viewing (**figure 2.4**). The nosepiece holds three to five **objectives** with lenses of differing magnifying power and can be rotated to position any objective beneath the body assembly. Ideally a microscope should

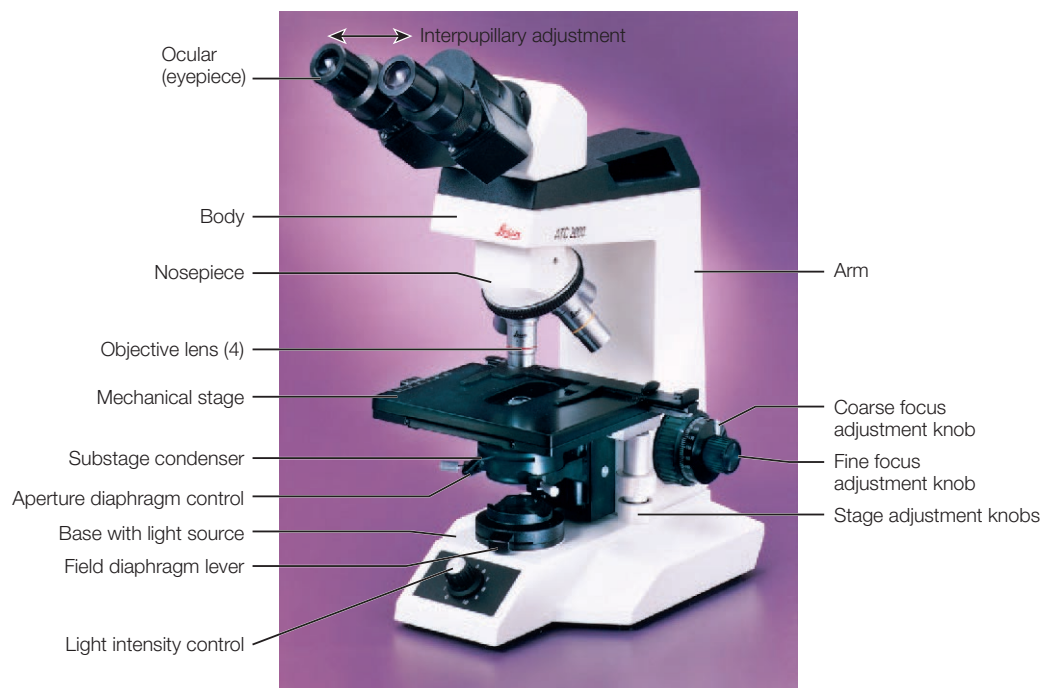


Figure 2.3 A Bright-Field Microscope. The parts of a modern bright-field microscope. The microscope pictured is somewhat more sophisticated than those found in many student laboratories. For example, it is binocular (has two eyepieces) and has a mechanical stage, an adjustable substage condenser, and a built-in illuminator.

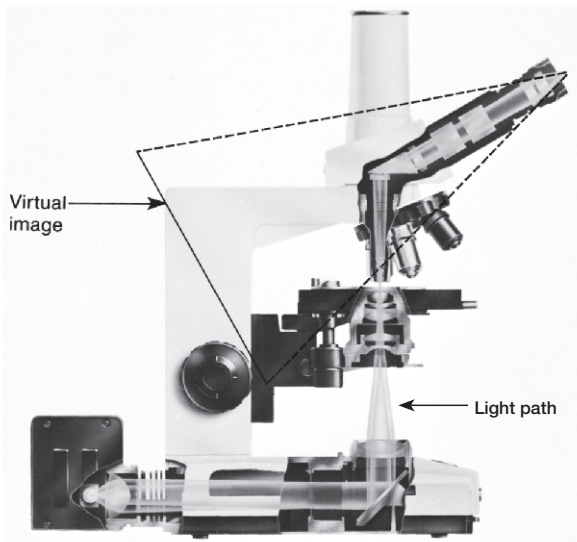


Figure 2.4 A Microscope's Light Path. The light path in an advanced bright-field microscope and the location of the virtual image are shown. (See also figure 2.23.)

be **parfocal**—that is, the image should remain in focus when objectives are changed.

The path of light through a bright-field microscope is shown in figure 2.4. The objective lens forms an enlarged real image within the microscope, and the eyepiece lens further magnifies this primary image. When one looks into a microscope, the enlarged specimen image, called the virtual image, appears to lie just beyond the stage about 25 cm away. The total magnification is calculated by multiplying the objective and eyepiece magnifications together. For example, if a 45 $\times$ objective is used with a 10 $\times$ eyepiece, the overall magnification of the specimen will be 450 $\times$.

Microscope Resolution

The most important part of the microscope is the objective, which must produce a clear image, not just a magnified one. Thus resolution is extremely important. **Resolution** is the ability of a lens to separate or distinguish between small objects that are close together. Much of the optical theory underlying microscope design was developed by the German physicist Ernst Abbé in the 1870s. The minimum distance (d) between two objects that reveals them as separate entities is given by the Abbé equation, in which lambda (λ) is the wavelength of light used to illuminate the specimen and $n \sin \theta$ is the numerical aperture (NA).

$$d = \frac{0.5\lambda}{n \sin \theta}$$

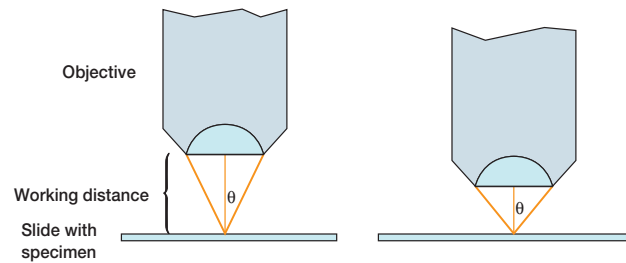


Figure 2.5 Numerical Aperture in Microscopy. The angular aperture θ is $\frac{1}{2}$ the angle of the cone of light that enters a lens from a specimen, and the numerical aperture is $n \sin \theta$. In the right-hand illustration the lens has larger angular and numerical apertures; its resolution is greater and its working distance smaller.

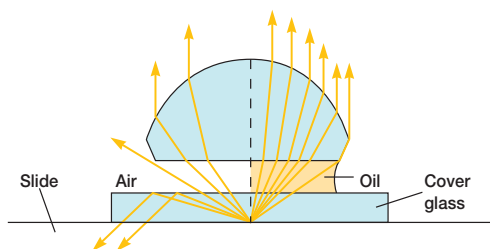
As d becomes smaller, the resolution increases, and finer detail can be discerned in a specimen.

The preceding equation indicates that a major factor in resolution is the wavelength of light used. The wavelength must be shorter than the distance between two objects or they will not be seen clearly. Thus the greatest resolution is obtained with light of the shortest wavelength, light at the blue end of the visible spectrum (in the range of 450 to 500 nm). [The electromagnetic spectrum of radiation \(p. 130\).](#)

The **numerical aperture** ($n \sin \theta$) is more difficult to understand. Theta is defined as $\frac{1}{2}$ the angle of the cone of light entering an objective (**figure 2.5**). Light that strikes the microorganism after passing through a condenser is cone-shaped. When this cone has a narrow angle and tapers to a sharp point, it does not spread out much after leaving the slide and therefore does not adequately separate images of closely packed objects. The resolution is low. If the cone of light has a very wide angle and spreads out rapidly after passing through a specimen, closely packed objects appear widely separated and are resolved. The angle of the cone of light that can enter a lens depends on the refractive index (n) of the medium in which the lens works, as well as upon the objective itself. The refractive index for air is 1.00. Since $\sin \theta$ cannot be greater than 1 (the maximum θ is 90° and $\sin 90^\circ$ is 1.00), no lens working in air can have a numerical aperture greater than 1.00. The only practical way to raise the numerical aperture above 1.00, and therefore achieve higher resolution, is to increase the refractive index with immersion oil, a colorless liquid with the same refractive index as glass (**table 2.2**). If air is replaced with immersion oil, many light rays that did not enter the objective due to reflection and refraction at the surfaces of the objective lens and slide will now do so (**figure 2.6**). An increase in numerical aperture and resolution results.

Table 2.2 The Properties of Microscope Objectives

Property	Objective			
	Scanning	Low Power	High Power	Oil Immersion
Magnification	4×	10×	40–45×	90–100×
Numerical aperture	0.10	0.25	0.55–0.65	1.25–1.4
Approximate focal length (<i>f</i>)	40 mm	16 mm	4 mm	1.8–2.0 mm
Working distance	17–20 mm	4–8 mm	0.5–0.7 mm	0.1 mm
Approximate resolving power with light of 450 nm (blue light)	2.3 μ m	0.9 μ m	0.35 μ m	0.18 μ m

**Figure 2.6** The Oil Immersion Objective. An oil immersion objective operating in air and with immersion oil.

The resolution of a microscope depends upon the numerical aperture of its condenser as well as that of the objective. This is evident from the equation describing the resolution of the complete microscope.

$$d_{\text{microscope}} = \frac{\lambda}{(\text{NA}_{\text{objective}} + \text{NA}_{\text{condenser}})}$$

Most microscopes have a condenser with a numerical aperture between 1.2 and 1.4. However, the condenser numerical aperture will not be much above about 0.9 unless the top of the condenser is oiled to the bottom of the slide. During routine microscope operation, the condenser usually is not oiled and this limits the overall resolution, even with an oil immersion objective.

The limits set on the resolution of a light microscope can be calculated using the Abbé equation. The maximum theoretical resolving power of a microscope with an oil immersion objective (numerical aperture of 1.25) and blue-green light is approximately 0.2 μ m.

$$d = \frac{(0.5)(530 \text{ nm})}{1.25} = 212 \text{ nm or } 0.2 \mu\text{m}$$

At best, a bright-field microscope can distinguish between two dots around 0.2 μ m apart (the same size as a very small bacterium).

Normally a microscope is equipped with three or four objectives ranging in magnifying power from 4×

to 100×

(table 2.2). The **working distance** of an objective is the distance between the front surface of the lens and the surface of the cover glass (if one is used) or the specimen when it is in sharp focus. Objectives with large numerical apertures and great resolving power have short working distances.

The Dark-Field Microscope

The largest useful magnification increases the size of the smallest resolvable object enough to be visible. Our eye can just detect a speck 0.2 mm in diameter, and consequently the useful limit of magnification is about 1,000 times the numerical aperture of the objective lens. Most standard microscopes come with 10×

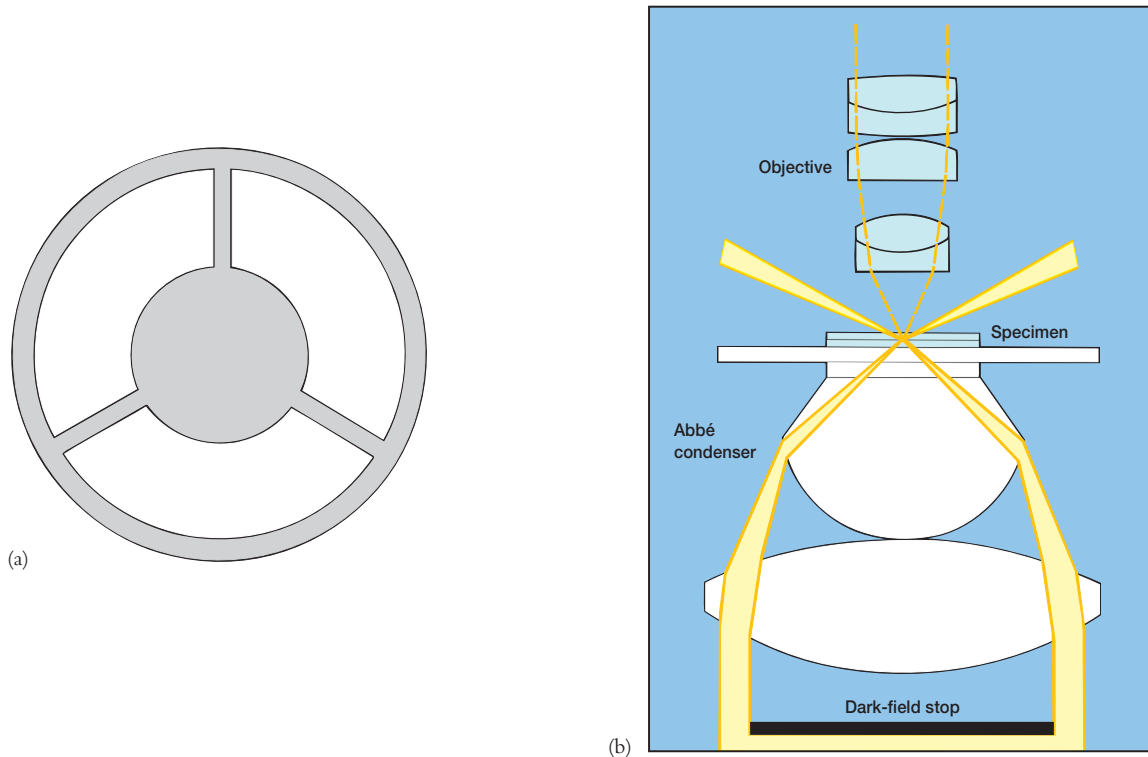


Figure 2.7 Dark-Field Microscopy. The simplest way to convert a microscope to dark-field microscopy is to place (a) a dark-field stop underneath (b) the condenser lens system. The condenser then produces a hollow cone of light so that the only light entering the objective comes from the specimen.

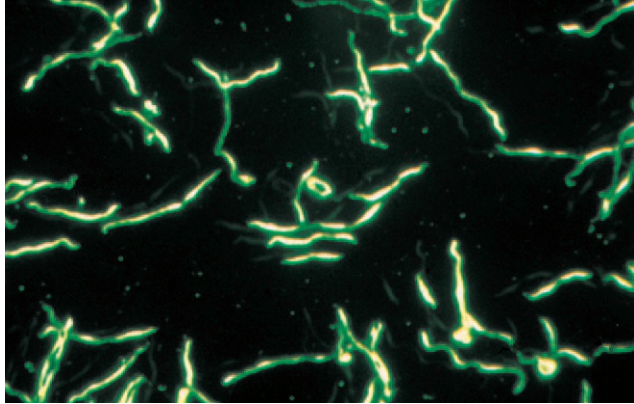
(figure 2.8a,b); because the background is dark, this type of microscopy is called **dark-field microscopy**. Considerable internal structure is often visible in larger eucaryotic microorganisms (figure 2.8b). The dark-field microscope is used to identify bacteria like the thin and distinctively shaped *Treponema pallidum* (figure 2.8a), the causative agent of syphilis.

The Phase-Contrast Microscope

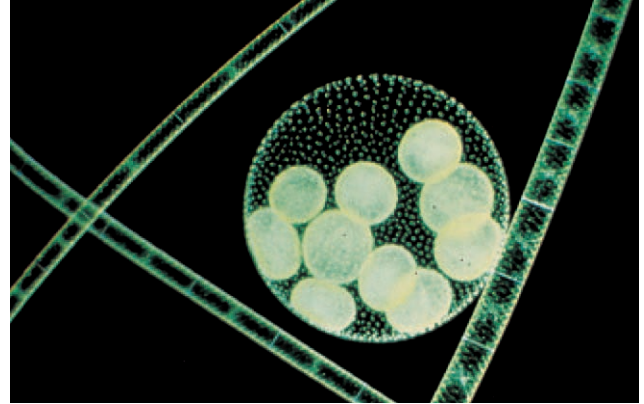
Unpigmented living cells are not clearly visible in the bright-field microscope because there is little difference in contrast between the cells and water. Thus microorganisms often must be fixed and stained before observation to increase contrast and create variations in color between cell structures. A **phase-contrast microscope** converts slight differences in re-

fractive index and cell density into easily detected variations in light intensity and is an excellent way to observe living cells (figure 2.8c–e).

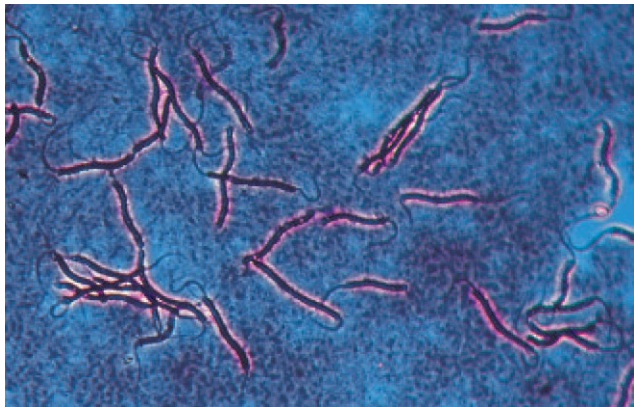
The condenser of a phase-contrast microscope has an annular stop, an opaque disk with a thin transparent ring, which produces a hollow cone of light (figure 2.9). As this cone passes through a cell, some light rays are bent due to variations in density and refractive index within the specimen and are retarded by about $\frac{1}{4}$ wavelength. The deviated light is focused to form an image of the object. Undeviated light rays strike a phase ring in the phase plate, a special optical disk located in the objective, while the deviated rays miss the ring and pass through the rest of the plate. If the phase ring is constructed in such a way that the undeviated light passing through it is advanced by $\frac{1}{4}$ wavelength, the deviated and undeviated waves will be about $\frac{1}{2}$ wavelength out of



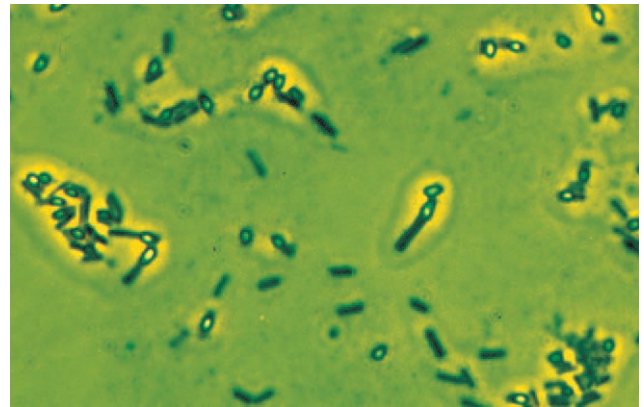
(a)



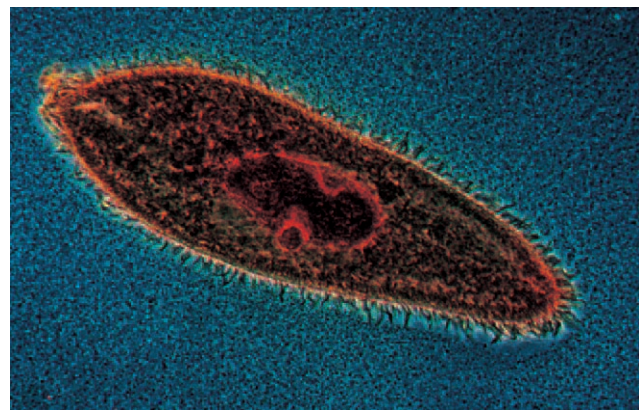
(b)



(c)



(d)



(e)

Figure 2.8 Examples of Dark-Field and Phase-Contrast Microscopy. (a) *Treponema pallidum*, the spirochete that causes syphilis; dark-field microscopy ($\times 500$). (b) *Volvox* and *Spirogyra*; dark-field microscopy ($\times 175$). Note daughter colonies within the mature *Volvox* colony (center) and the spiral chloroplasts of *Spirogyra* (left and right). (c) *Spirillum volutans*, a very large bacterium with flagellar bundles; phase-contrast microscopy ($\times 210$). (d) *Clostridium botulinum*, the bacterium responsible for botulism, with subterminal oval endospores; phase-contrast microscopy ($\times 600$). (e) *Paramecium* stained to show a large central macronucleus with a small spherical micronucleus at its side; phase-contrast microscopy ($\times 100$).

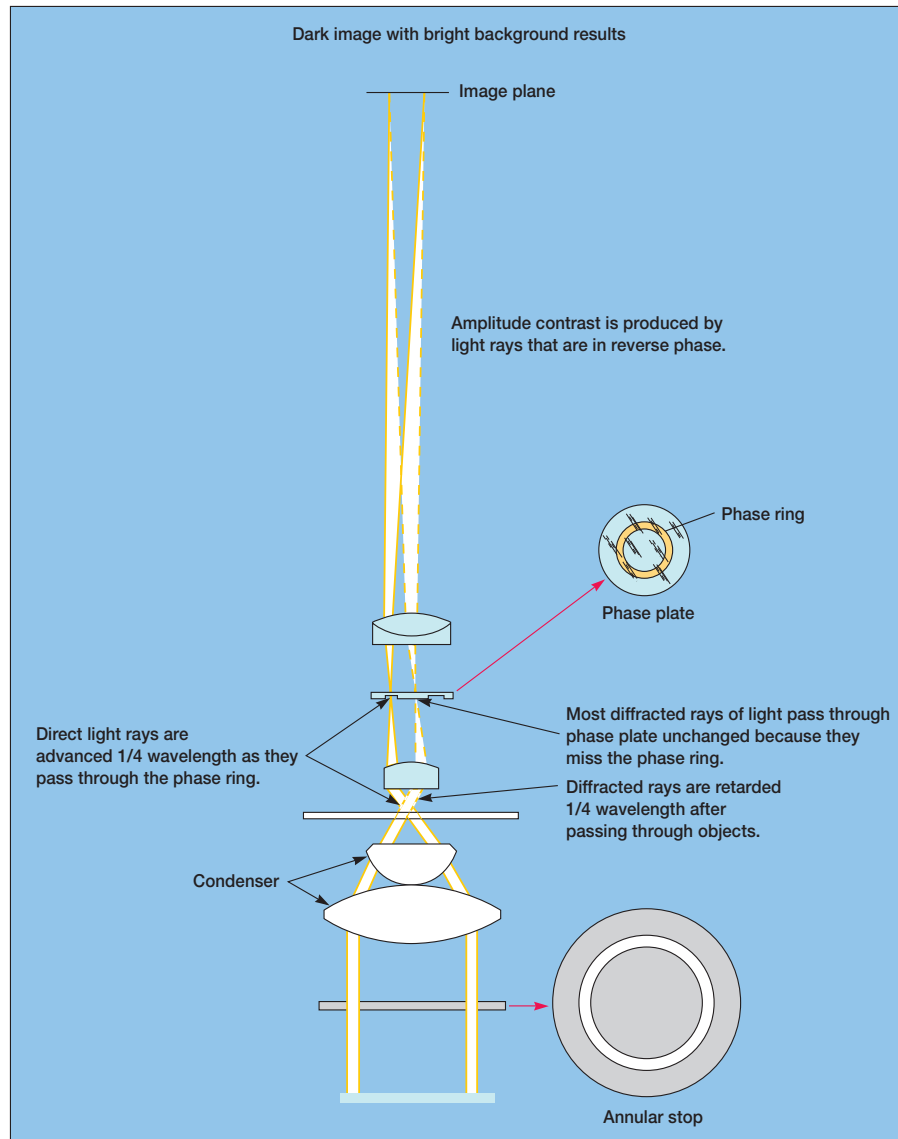


Figure 2.9 Phase-Contrast Microscopy. The optics of a dark-phase-contrast microscope.

phase and will cancel each other when they come together to form an image (**figure 2.10**). The background, formed by undeviated light, is bright, while the unstained object appears dark and well-defined. This type of microscopy is called **dark-phase-contrast microscopy**. Color filters often are used to improve the image (**figure 2.8c,d**).

Phase-contrast microscopy is especially useful for studying microbial motility, determining the shape of living cells,

and detecting bacterial components such as endospores and inclusion bodies that contain poly- β -hydroxybutyrate, poly-metaphosphate, sulfur, or other substances (*see chapter 3*). These are clearly visible (**figure 2.8d**) because they have refractive indexes markedly different from that of water. Phase-contrast microscopes also are widely used in studying eucaryotic cells.

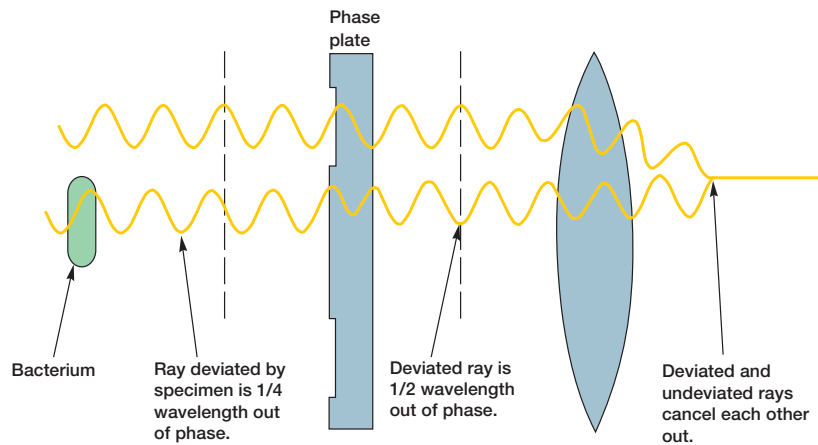


Figure 2.10 The Production of Contrast in Phase Microscopy. The behavior of deviated and undeviated or undiffracted light rays in the dark-phase-contrast microscope. Because the light rays tend to cancel each other out, the image of the specimen will be dark against a brighter background.

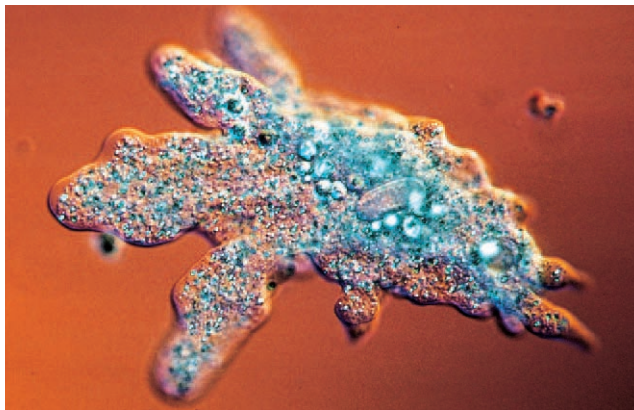


Figure 2.11 Differential Interference Contrast Microscopy. A micrograph of the protozoan *Amoeba proteus*. The three-dimensional image contains considerable detail and is artificially colored ($\times 160$).

The Differential Interference Contrast Microscope

The **differential interference contrast (DIC) microscope** is similar to the phase-contrast microscope in that it creates an image by detecting differences in refractive indices and thickness. Two beams of plane polarized light at right angles to each other are generated by prisms. In one design, the object beam passes through the specimen, while the reference beam passes through a clear area of the

slide. After passing through the specimen, the two beams are combined and interfere with each other to form an image. A live, unstained specimen appears brightly colored and three-dimensional (**figure 2.11**). Structures such as cell walls, endospores, granules, vacuoles, and eucaryotic nuclei are clearly visible.

The Fluorescence Microscope

The microscopes thus far considered produce an image from light that passes through a specimen. An object also can be seen because it actually emits light, and this is the basis of fluorescence microscopy. When some molecules absorb radiant energy, they become excited and later release much of their trapped energy as light. Any light emitted by an excited molecule will have a longer wavelength (or be of lower energy) than the radiation originally absorbed. **Fluorescent light** is emitted very quickly by the excited molecule as it gives up its trapped energy and returns to a more stable state.

The **fluorescence microscope** (**figure 2.12**) exposes a specimen to ultraviolet, violet, or blue light and forms an image of the object with the resulting fluorescent light. A mercury vapor arc lamp or other source produces an intense beam, and heat transfer is limited by a special infrared filter. The light passes through an exciter filter that transmits only the desired wavelength. A dark-field condenser provides a black background against which the fluorescent objects glow. Usually the specimens have been stained with dye molecules, called **fluorochromes**, that fluoresce brightly upon exposure to light of a specific wavelength, but some microorganisms are autofluorescing. The microscope forms an image of the fluorochrome-labeled microorganisms

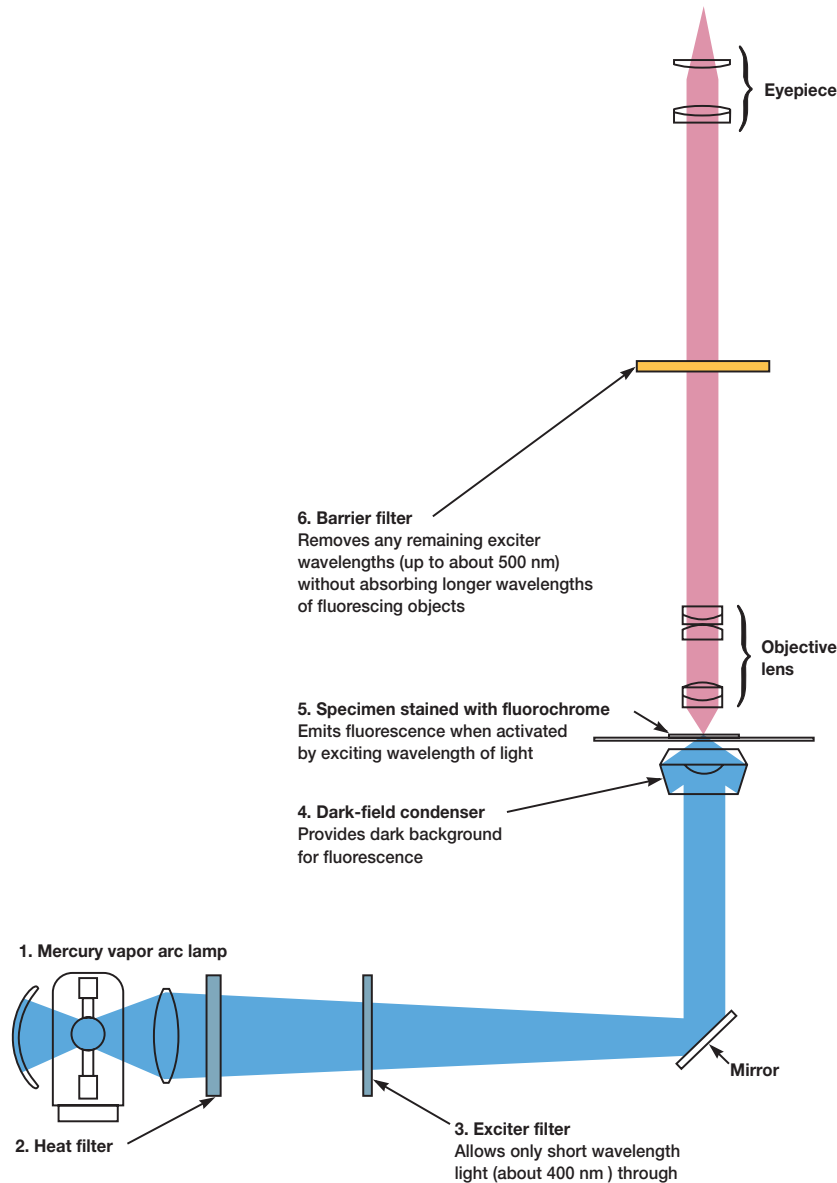


Figure 2.12 Fluorescence Microscopy. The principles of operation of a fluorescence microscope.

from the light emitted when they fluoresce (**figure 2.13**). A barrier filter positioned after the objective lenses removes any remaining ultraviolet light, which could damage the viewer's eyes, or blue and violet light, which would reduce the image's contrast.

The fluorescence microscope has become an essential tool in medical microbiology and microbial ecology. Bacterial pathogens (e.g., *Mycobacterium tuberculosis*, the cause of tuberculosis) can be identified after staining them with fluorochromes or specifically labeling them with fluorescent antibodies using immunofluorescence procedures. In ecological

studies the fluorescence microscope is used to observe microorganisms stained with fluorochrome-labeled probes or fluorochromes such as acridine orange and DAPI (diamidino-2-phenylindole, a DNA-specific stain). The stained organisms will fluoresce orange or green and can be detected even in the midst of other particulate material. It is even possible to distinguish live bacteria from dead bacteria by the color they fluoresce after treatment with a special mixture of stains (**figure 2.13d**). Thus the microorganisms can be viewed and directly counted in a relatively undisturbed ecological niche. [Immunofluorescence and diagnostic microbiology](#) (pp. 781, 831–32).

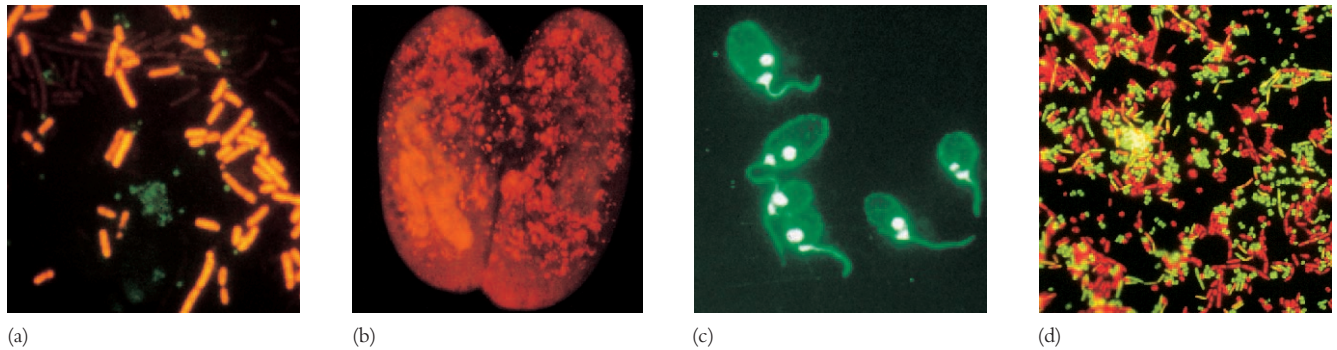


Figure 2.13 Examples of Fluorescence Microscopy. (a) *Escherichia coli* stained with fluorescent antibodies ($\times 600$). The green material is debris. (b) *Paramecium tetraurelia* conjugating; acridine-orange fluorescence ($\times 125$). (c) The flagellate protozoan *Crithidia lucilliae* stained with fluorescent antibodies to show the kinetoplast ($\times 1,000$). (d) A mixture of *Micrococcus luteus* and *Bacillus cereus* (the rods). The live bacteria fluoresce green; dead bacteria are red.

1. List the parts of a light microscope and their functions.
2. Define resolution, numerical aperture, working distance, and fluorochrome.
3. How does resolution depend upon the wavelength of light, refractive index, and the numerical aperture? What are the functions of immersion oil and the substage condenser?
4. Briefly describe how dark-field, phase-contrast, differential interference contrast, and fluorescence microscopes work and the kind of image provided by each. Give a specific use for each type.

2.3 Preparation and Staining of Specimens

Although living microorganisms can be directly examined with the light microscope, they often must be fixed and stained to increase visibility, accentuate specific morphological features, and preserve them for future study.

Fixation

The stained cells seen in a microscope should resemble living cells as closely as possible. **Fixation** is the process by which the internal and external structures of cells and microorganisms are preserved and fixed in position. It inactivates enzymes that might disrupt cell morphology and toughens cell structures so that they do not change during staining and observation. A microorganism usually is killed and attached firmly to the microscope slide during fixation.

There are two fundamentally different types of fixation. (1) Bacteriologists heat-fix bacterial smears by gently flame heating an air-dried film of bacteria. This adequately preserves overall morphology but not structures within cells. (2) Chemical fixation must be used to protect fine cellular substructure and the morphol-

ogy of larger, more delicate microorganisms. Chemical fixatives penetrate cells and react with cellular components, usually proteins and lipids, to render them inactive, insoluble, and immobile. Common fixative mixtures contain such components as ethanol, acetic acid, mercuric chloride, formaldehyde, and glutaraldehyde.

Dyes and Simple Staining

The many types of dyes used to stain microorganisms have two features in common. (1) They have **chromophore groups**, groups with conjugated double bonds that give the dye its color. (2) They can bind with cells by ionic, covalent, or hydrophobic bonding. For example, a positively charged dye binds to negatively charged structures on the cell.

Ionizable dyes may be divided into two general classes based on the nature of their charged group.

1. **Basic dyes**—methylene blue, basic fuchsin, crystal violet, safranin, malachite green—have positively charged groups (usually some form of pentavalent nitrogen) and are generally sold as chloride salts. Basic dyes bind to negatively charged molecules like nucleic acids and many proteins. Because the surfaces of bacterial cells also are negatively charged, basic dyes are most often used in bacteriology.
2. **Acid dyes**—eosin, rose bengal, and acid fuchsin—possess negatively charged groups such as carboxyls ($-\text{COOH}$) and phenolic hydroxyls ($-\text{OH}$). Acid dyes, because of their negative charge, bind to positively charged cell structures.

The pH may alter staining effectiveness since the nature and degree of the charge on cell components change with pH. Thus anionic dyes stain best under acidic conditions when proteins and many other molecules carry a positive charge; basic dyes are most effective at higher pHs.

Although ionic interactions are probably the most common means of attachment, dyes also bind through covalent bonds or because of their solubility characteristics. For instance, DNA can be stained by the Feulgen procedure in which Schiff's reagent is covalently attached to its deoxyribose sugars after hydrochloric acid treatment. Sudan III (Sudan Black) selectively stains lipids because it is lipid soluble but will not dissolve in aqueous portions of the cell.

Microorganisms often can be stained very satisfactorily by **simple staining**, in which a single staining agent is used. Simple staining's value lies in its simplicity and ease of use. One covers the fixed smear with stain for the proper length of time, washes the excess stain off with water, and blots the slide dry. Basic dyes like crystal violet, methylene blue, and carbolfuchsin are frequently used to determine the size, shape, and arrangement of bacteria.

Differential Staining

Differential staining procedures divide bacteria into separate groups based on staining properties. The **Gram stain**, developed in 1884 by the Danish physician Christian Gram, is the most widely employed staining method in bacteriology. It is a differential staining procedure because it divides bacteria into two classes—gram negative and gram positive. [Gram-positive and gram-negative bacteria](#) (pp. 55–60, 440–41).

In the first step of the Gram-staining procedure (**figure 2.14**), the smear is stained with the basic dye crystal violet, the primary stain. It is followed by treatment with an iodine solution functioning as a **mordant**. That is, the iodine increases the interaction between the cell and the dye so that the cell is stained more strongly. The smear is next decolorized by washing with ethanol or acetone. This step generates the differential aspect of the Gram stain; gram-positive bacteria retain the crystal violet, whereas gram-negative bacteria lose their crystal violet and become colorless. Finally, the smear is counterstained with a simple, basic dye different in color from crystal violet. Safranin, the most common counterstain, colors gram-negative bacteria pink to red and leaves gram-positive bacteria dark purple (**figure 2.15**). [Cell wall structure and the mechanism of the Gram stain](#) (p. 60).

Acid-fast staining is another important differential staining procedure. A few species, particularly those in the genus *Mycobacterium* (see chapter 24) do not bind simple stains readily and must be stained by a harsher treatment: heating with a mixture of basic fuchsin and phenol (the Ziehl-Neelsen method). Once basic fuchsin has penetrated with the aid of heat and phenol, acid-fast cells are not easily decolorized by an acid-alcohol wash and hence remain red. This is due to the quite high lipid content of acid-fast cell walls; in particular, mycolic acid—a group of branched chain hydroxy lipids—appears responsible for acid-fastness. Non-acid-fast bacteria are decolorized by acid-alcohol and thus are stained blue by methylene blue counterstain. This method is used to identify *Mycobacterium tuberculosis* and *M. leprae* (**figure 2.16**), the pathogens responsible for tuberculosis and leprosy, respectively.

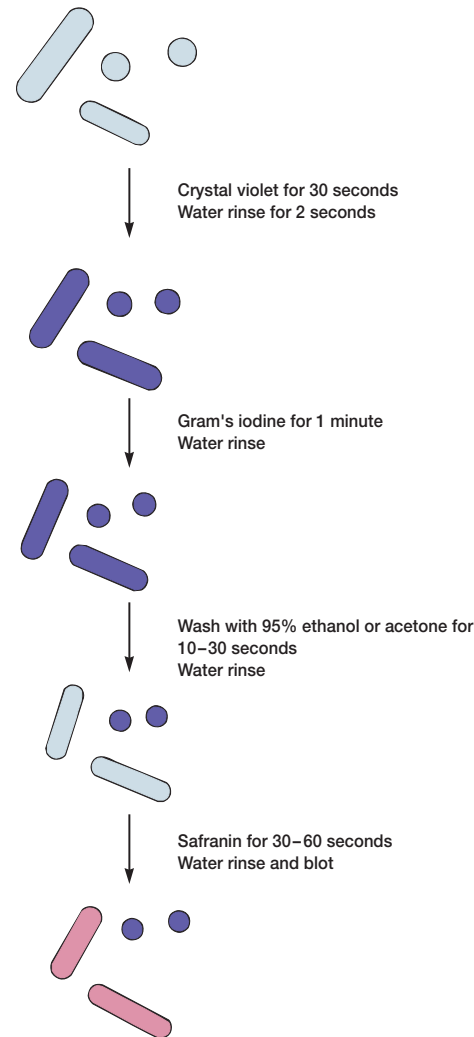
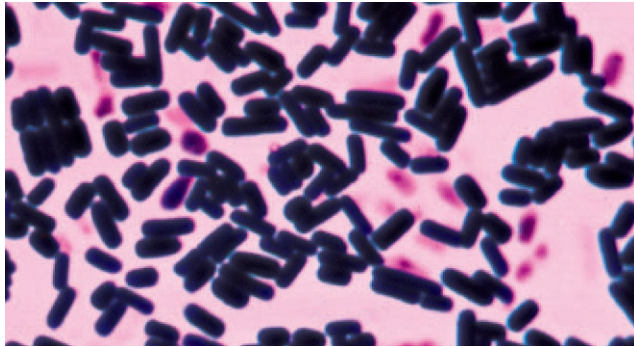


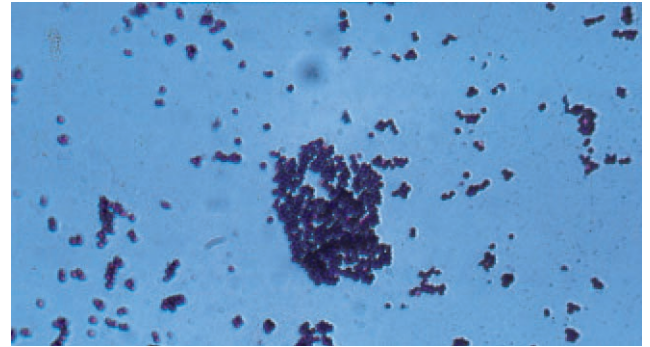
Figure 2.14 The Gram-Staining Procedure. Note that decolorization with ethanol or acetone removes crystal violet from gram-negative cells but not from gram-positive cells. The gram-negative cells then turn pink to red when counterstained with safranin.

Staining Specific Structures

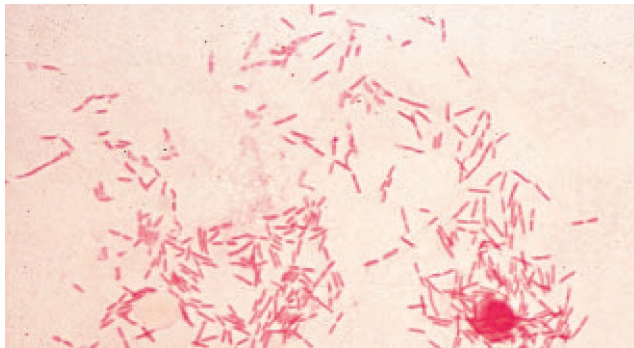
Many special staining procedures have been developed over the years to study specific bacterial structures with the light microscope. One of the simplest is **negative staining**, a technique that reveals the presence of the diffuse capsules surrounding many bacteria. Bacteria are mixed with India ink or Nigrosin dye and spread out in a thin film on a slide. After air-drying, bacteria appear as lighter bodies in the midst of a blue-black background because ink and dye particles cannot penetrate either the bacterial cell or its capsule. The extent of the light region is determined by the size of the capsule and of the cell itself. There is little distortion of bacterial shape, and the cell can be counterstained for even greater visibility (**figure 2.17**). [Capsules and slime layers](#) (pp. 61–62).



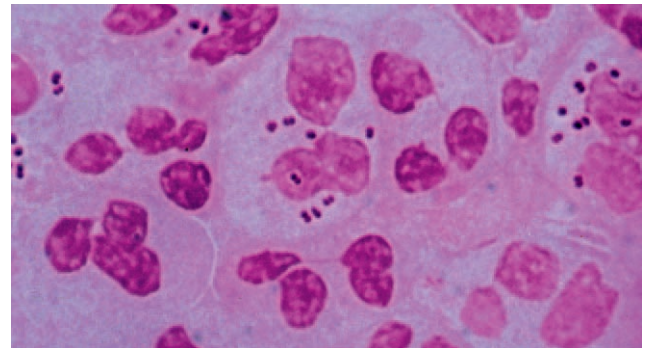
(a)



(b)



(c)



(d)

Figure 2.15 Examples of Gram Staining. (a) Gram-positive *Clostridium perfringens* ($\times 800$). Some rods have stained pink rather than purple, as often happens when gram-positive cells age. (b) *Staphylococcus aureus*. Gram stain, bright-field microscopy ($\times 1,000$). The gram-positive cocci associate in grapelike clusters. (c) *Escherichia coli*, Gram stain ($\times 500$). (d) *Neisseria gonorrhoeae*. The diplococci are often within white blood cells ($\times 1,000$).

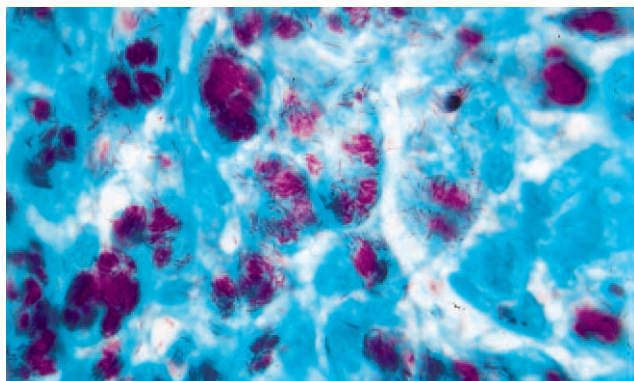


Figure 2.16 Acid-Fast Staining. *Mycobacterium leprae*. Acid-fast stain ($\times 380$). Note the masses of red bacteria within host cells.

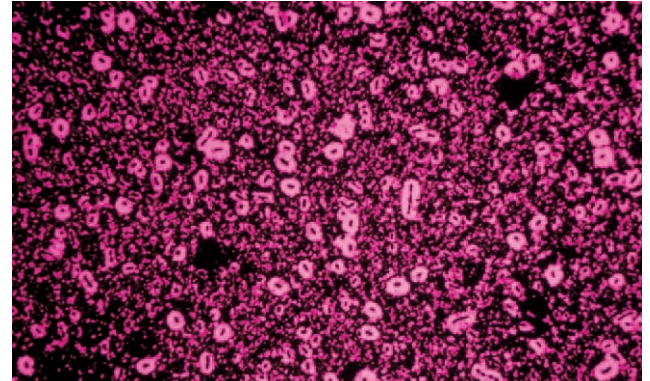


Figure 2.17 Negative Staining. *Klebsiella pneumoniae* negatively stained with India ink to show its capsules ($\times 900$).

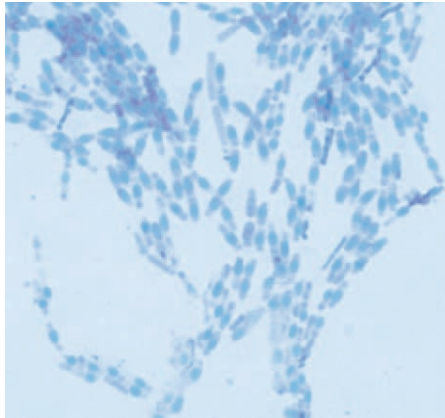


Figure 2.18 Spore Staining. *Bacillus cereus* stained with the Schaeffer-Fulton procedure. Note the central, elliptical blue to green spores within the red to purple cells ($\times 1,000$).

Bacteria in the genera *Bacillus* and *Clostridium* (see chapter 23) form an exceptionally resistant structure capable of surviving for long periods in an unfavorable environment. This dormant structure is called an endospore since it develops within the cell. Endospore morphology and location vary with species and often are valuable in identification; endospores may be spherical to elliptical and either smaller or larger than the diameter of the parent bacterium. They can be observed with the phase-contrast microscope or negative staining. Endospores are not stained well by most dyes, but once stained, they strongly resist decolorization. This property is the basis of most **spore staining** methods (**figure 2.18**). In the Schaeffer-Fulton procedure, endospores are first stained by heating bacteria with malachite green, which is a very strong stain that can penetrate endospores. After malachite green treatment, the rest of the cell is washed free of dye with water and is counterstained with safranin. This technique yields a green endospore resting in a pink to red cell. *Bacterial endospore structure* (pp. 68–71).

Bacterial flagella are fine, threadlike organelles of locomotion that are so slender (about 10 to 30 nm in diameter) they can only be seen directly using the electron microscope. To observe them with the light microscope, the thickness of flagella is increased by coating them with mordants like tannic acid and potassium alum, and they are stained with pararosaniline (Leifson method) or basic fuchsin (Gray method). **Flagella staining** procedures provide taxonomically valuable information about the presence and distribution pattern of flagella (**figure 2.19**; see also **figure 3.31**). *The bacterial flagellum* (pp. 63–66).



Figure 2.19 Example of Flagella Staining. *Spirillum volutans* with bipolar tufts of flagella ($\times 400$). (See also **figure 3.31**.)

2.4 Electron Microscopy

For centuries the light microscope has been the most important instrument for studying microorganisms. The electron microscope now has transformed microbiology and added immeasurably to our knowledge. The nature of the electron microscope and the ways in which specimens are prepared for observation are reviewed briefly in this section.

The Transmission Electron Microscope

The very best light microscope has a resolution limit of about 0.2 μm . Because bacteria usually are around 1 μm in diameter, only their general shape and major morphological features are visible in the light microscope. The detailed internal structure of larger microorganisms also cannot be effectively studied by light microscopy. These limitations arise from the nature of visible light waves, not from any inadequacy of the light microscope itself.

Recall that the resolution of a light microscope increases with a decrease in the wavelength of the light it uses for illumination. Electron beams behave like radiation and can be focused much as light is in a light microscope. If electrons illuminate the specimen, the microscope's resolution is enormously increased because the wavelength of the radiation is around 0.005 nm, approximately 100,000 times shorter than that of visible light. The transmission electron microscope has a practical resolution roughly 1,000 times better than the light microscope; with many electron microscopes, points closer than 5 Å or 0.5 nm can be distinguished, and the useful magnification is well over 100,000 $\times$ (**figure 2.20**). The value of the electron microscope is evident on comparison of the photographs in **figure 2.21**; microbial morphology can now be studied in great detail.

A modern **transmission electron microscope (TEM)** is complex and sophisticated (**figure 2.22**), but the basic principles

1. Define fixation, dye, chromophore, basic dye, acid dye, simple staining, differential staining, mordant, negative staining, and acid-fast staining.
2. Describe the Gram-stain procedure and how it works.
3. How would you visualize capsules, endospores, and flagella?

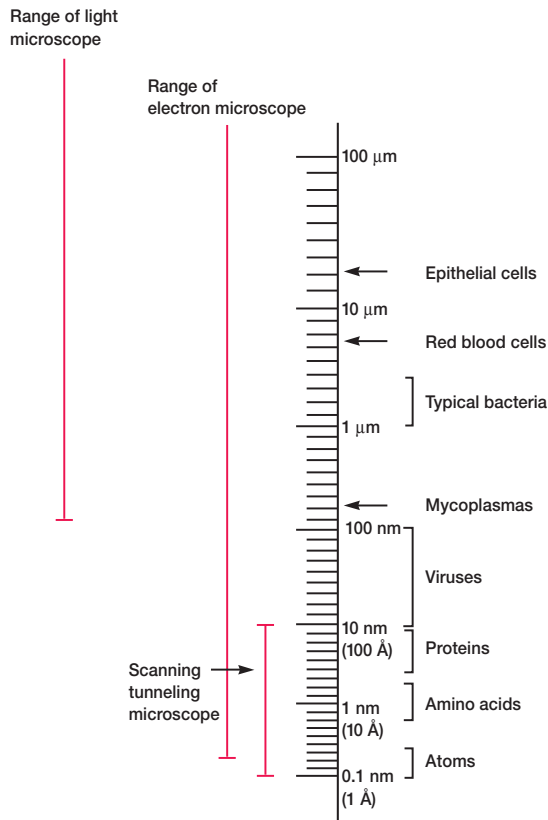


Figure 2.20 The Limits of Microscopic Resolution. Dimensions are indicated with a logarithmic scale (each major division represents a tenfold change in size). To the right side of the scale are the approximate sizes of cells, bacteria, viruses, molecules, and atoms.



Figure 2.22 A Modern Transmission Electron Microscope. The electron gun is at the top of the central column, and the magnetic lenses are within the column. The image on the fluorescent screen may be viewed through a magnifier positioned over the viewing window. The camera is in a compartment below the screen.

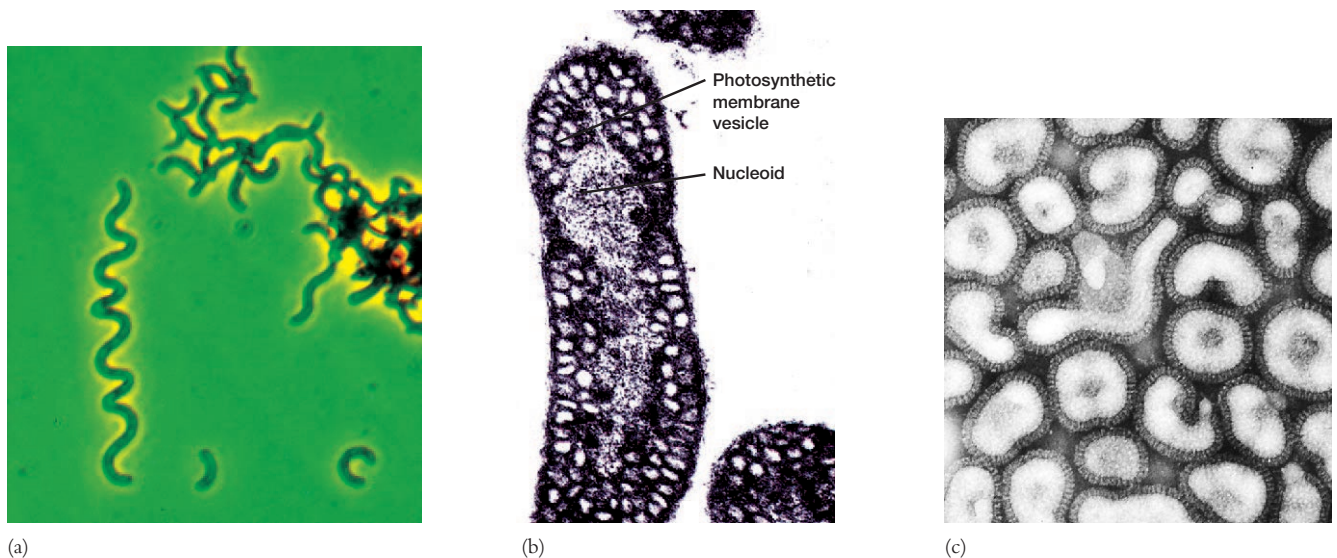


Figure 2.21 Light and Electron Microscopy. A comparison of light and electron microscopic resolution. (a) *Rhodospirillum rubrum* in phase-contrast light microscope (×600). (b) A thin section of *R. rubrum* in transmission electron microscope (×100,000). (c) A micrograph of human influenza viruses (×282,000). The particles are about 100 nm in diameter, much smaller than bacterial cells.

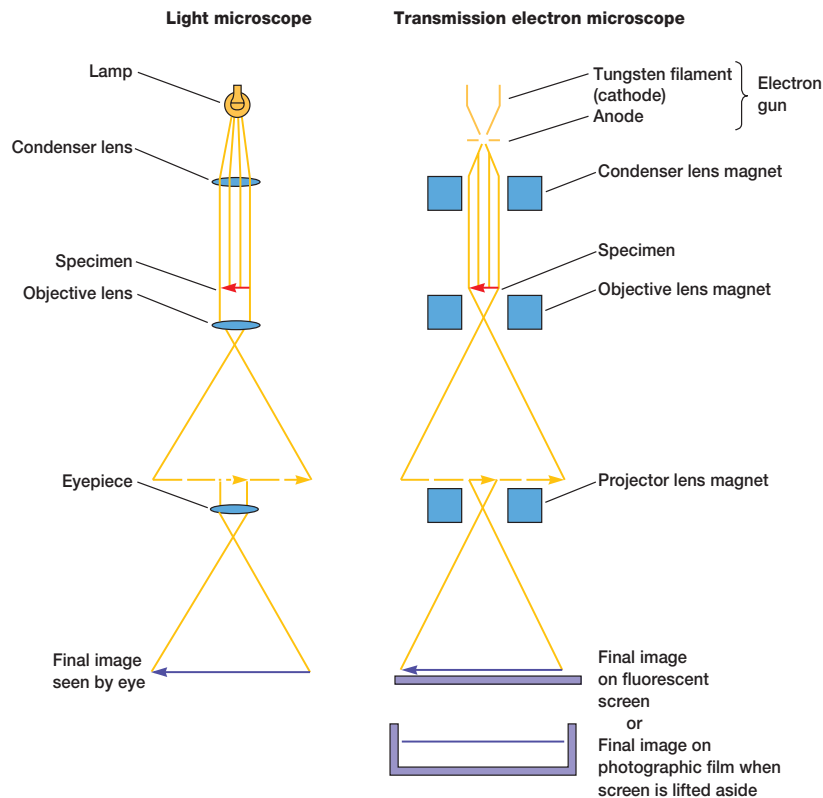


Figure 2.23 Transmission Electron Microscope Operation. An overview of TEM operation and a comparison of the operation of light and transmission electron microscopes.

behind its operation can be understood readily. A heated tungsten filament in the electron gun generates a beam of electrons that is then focused on the specimen by the condenser (**figure 2.23**). Since electrons cannot pass through a glass lens, doughnut-shaped electromagnets called magnetic lenses are used to focus the beam. The column containing the lenses and specimen must be under high vacuum to obtain a clear image because electrons are deflected by collisions with air molecules. The specimen scatters electrons passing through it, and the beam is focused by magnetic lenses to form an enlarged, visible image of the specimen on a fluorescent screen. A denser region in the specimen scatters more electrons and therefore appears darker in the image since fewer electrons strike that area of the screen. In contrast, electron-transparent regions are brighter. The screen can also be moved aside and the image captured on photographic film as a permanent record.

Specimen Preparation

Table 2.3 compares some of the important features of light and electron microscopes. The distinctive features of the TEM place harsh restrictions on the nature of samples that can be viewed and the means by which those samples must be prepared. Since electrons are quite easily absorbed and scattered by solid matter, only

extremely thin slices of a microbial specimen can be viewed in the average TEM. The specimen must be around 20 to 100 nm thick, about $\frac{1}{50}$ to $\frac{1}{100}$ the diameter of a typical bacterium, and able to maintain its structure when bombarded with electrons under a high vacuum! Such a thin slice cannot be cut unless the specimen has support of some kind; the necessary support is provided by plastic. After fixation with chemicals like glutaraldehyde or osmium tetroxide to stabilize cell structure, the specimen is dehydrated with organic solvents (e.g., acetone or ethanol). Complete dehydration is essential because most plastics used for embedding are not water soluble. Next the specimen is soaked in unpolymerized, liquid epoxy plastic until it is completely permeated, and then the plastic is hardened to form a solid block. Thin sections are cut from this block with a glass or diamond knife using a special instrument called an ultramicrotome.

Cells usually must be stained before they can be seen clearly in the bright-field microscope; the same is true for observations with the TEM. The probability of electron scattering is determined by the density (atomic number) of the specimen atoms. Biological molecules are composed primarily of atoms with low atomic numbers (H, C, N, and O), and electron scattering is fairly constant throughout the unstained cell. Therefore specimens are prepared for observation by soaking